

IO SKY PLATFORM

Master Software Requirements Specification

Version 1.1

RELEASE CANDIDATE — DEVELOPER BASELINE

Approval Status: PENDING RESOLUTION OF OPEN PRODUCT OWNER DECISIONS

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Module Index

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Module 3	AI Scan Platform
Module 4	Client Portal
Module 5	Developer Portal
Module 6	Admin Portal
Module 7	Super Admin Portal
Module 8	CRM & Sales Management
Module 9	Project & Delivery Management
Module 10	Commercial Billing & Subscription Management
Module 11	Notifications & Communication
Module 12	File & Document Management
Module 13	AI Intelligence Layer
Module 14	Audit, Compliance & Security Monitoring
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Module 17	Workflow & Business Process Management
Module 18	Platform Configuration & System Administration
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Appendix Index

Supplementary reference material supporting implementation and interpretation.

Appendix	Title
Appendix A	Glossary of Terms
Appendix B	Business Rule Index
Appendix C	Product Principle Index
Appendix D	Open Product Owner Decisions

CHAPTER 1

1. Executive Summary

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and business requirements for the IO SKY Platform.

It serves as the single source of truth for all stakeholders involved in the design, development, testing, deployment, and maintenance of the platform.

The purpose of this document is to clearly define what the platform must do, while leaving all implementation decisions—including architecture, database design, frameworks, and coding practices—to the development team.

This document is intentionally written from a

Product Owner perspective and focuses exclusively on business behaviour, user experience, workflows, permissions, and operational requirements.

1.2 Platform Overview

IO SKY is an Operational Intelligence Platform designed to help organizations improve operational performance, strengthen cybersecurity, automate business processes, and make better strategic decisions through AI-assisted analysis.

The platform combines intelligent assessments, operational dashboards, project delivery, customer collaboration, AI-assisted decision support, and administrative management into one integrated ecosystem.

Rather than offering isolated software tools, IO

SKY provides a connected operational infrastructure where every component contributes to a single objective:

Helping organizations operate more efficiently, securely, predictably, and at scale.

1.3 Primary Objectives

The platform shall enable organizations to:

- Assess operational maturity.
- Assess cybersecurity posture.
- Identify automation opportunities.

- Receive AI-assisted business insights.
- Collaborate with IO SKY throughout implementation projects.
- Track project progress.
- Access reports and documentation.
- Manage organizational communication through a centralized platform.

For IO SKY, the platform shall provide centralized operational control over customers, projects, AI

Scans, reporting, sales, billing, developers, and platform administration.

1.4 Scope

Version 1 of the platform includes:

- Public Website
- Identity & Authentication
- AI Scan Platform
- Client Portal
- Developer Portal
- Admin Portal
- Super Admin Portal
- CRM & Sales Management
- Project & Delivery Management
- Reporting
- Notifications
- Billing
- AI Intelligence Layer
- Audit Logging
- Analytics
- Platform Configuration
- Integrations

Features explicitly marked as future enhancements are considered out of scope for

Version 1.

1.5 Guiding Principle

The IO SKY Platform is not designed to function as a collection of independent modules.

Every module must operate as part of a single integrated ecosystem where information flows securely, consistently, and automatically between all authorized components.

No feature shall be developed in isolation if integration with existing platform functionality is required.

CHAPTER 2

2. Product Vision

2.1 Vision Statement

IO SKY exists to help organizations replace fragmented operations with connected operational intelligence.

Instead of relying on disconnected software solutions, manual processes, and isolated automation tools, organizations use IO SKY to gain centralized visibility, operational control, AI-assisted decision support, and scalable digital infrastructure.

The platform is designed to become the operational control center for every organization that uses it.

2.2 Positioning

IO SKY is not:

- A software development agency.
- A traditional AI agency.
- A marketing automation platform.
- A CRM provider.
- A cybersecurity consultancy.

IO SKY is an Operational Intelligence Platform that combines operational analysis, cybersecurity assessment, AI-assisted intelligence, project delivery, and customer collaboration into one unified platform.

2.3 Product Philosophy

Every feature developed within the platform must contribute to one or more of the following principles:

- Increase operational clarity.
- Reduce manual work.
- Improve decision-making.
- Strengthen organizational security.
- Standardize operational processes.
- Support sustainable business growth.
- Maintain enterprise-grade reliability.
- Preserve data integrity and organizational isolation.

Features that do not contribute to these objectives should not be included in the platform.

CHAPTER 3

3. Product Principles

The following principles apply to every module within the platform.

PP-001 Security by Default

Security is the default behaviour of the platform.

Every feature must follow the principle of least privilege.

PP-002 Human Oversight

Artificial Intelligence may generate recommendations, summaries, analyses, and draft content.

Artificial Intelligence shall never perform irreversible business actions without explicit human authorization unless explicitly configured by the Super Admin.

PP-003 Organization Isolation

Every organization operates within a completely isolated environment.

No user may access data belonging to another organization under any circumstance.

PP-004 Single Source of Truth

Business information must exist in one authoritative location only.

Duplicate business data should be avoided wherever possible.

PP-005 Role-Based Access Control

Access to every feature, module, and dataset shall be determined by assigned roles and permissions.

Users never manually select which portal to access after authentication.

PP-006 Complete Auditability

Every security-sensitive or business-critical action performed within the platform must be recorded in an immutable audit log.

PP-007 Enterprise First

The platform shall be designed for organizations with multiple departments, multiple users, and future scalability without requiring architectural redesign.

PP-008 Consistency

User experience, terminology, navigation, permissions, workflows, and visual behaviour must remain consistent across all modules.

CHAPTER 4

4. Global Business Rules

Purpose

This section defines the mandatory business rules that apply across the entire IO SKY

Platform.

These rules are platform-wide requirements and take precedence over individual module specifications. Every module, workflow, integration, AI process, and user interaction must comply with these business rules.

Authentication & Identity

BR-001 Authentication Required

Only authenticated users may access protected platform resources.

Public users may only access publicly available website pages.

BR-002 Single Identity

Each user shall have only one active account within the platform.

Multiple accounts for the same individual within the same Organization are not permitted.

BR-003 Automatic Portal Routing

Users shall never manually select a portal after authentication.

Following successful authentication, the platform shall automatically redirect the user to the appropriate portal based on their assigned role and permissions.

BR-004 Invitation-Based Access

Public account registration is not permitted.

User accounts may only be created through approved platform workflows, including:

- AI Scan purchase
- Admin invitation
- Super Admin invitation
- Approved customer onboarding
- Organization & Multi-Tenancy

BR-005 Organization Ownership

Every user must belong to at least one

Organization.

No user may exist without an associated

Organization unless explicitly designated as a platform-level user (e.g., Super Admin).

BR-006 Organization Isolation

Organizations must remain completely isolated.

Users shall never be able to access information belonging to another Organization.

This applies to all platform data, including projects, reports, invoices, AI Scans, files, messages, and notifications.

BR-007 Data Ownership

All business records shall belong to exactly one

Organization.

This includes, but is not limited to:

- AI Scans
- Reports
- Projects
- Files
- Messages
- Invoices
- Notifications
- Tasks
- CRM records

AI Business Rules

BR-008 AI Supports Human Decision-Making

Artificial Intelligence is designed to assist users by generating recommendations, analyses, summaries, and draft content.

AI shall not replace human decision-making for business-critical actions.

BR-009 Human Approval Required

Any AI-generated report intended for customer use must be reviewed and explicitly approved by an authorized IO SKY reviewer before publication.

AI-generated reports shall never be published automatically.

BR-010 AI Transparency

Whenever content has been generated or substantially modified by AI, the platform shall retain metadata indicating its origin for auditing purposes.

Communication

BR-011 Customer Communication

Clients shall never communicate directly with

Developers through the platform.

All customer communication shall be managed by Admins or Super Admins.

BR-012 Internal Communication

Developers may communicate internally with

Admins, Super Admins, and authorized

Technical Operators only.

Security

BR-013 Least Privilege

Every user shall receive only the minimum permissions required to perform their responsibilities.

BR-014 Multi-Factor Authentication

Multi-Factor Authentication (MFA) shall be mandatory for:

- Super Admin
- Admin
- Technical Operator
- Developers are strongly encouraged to enable
- MFA.

Clients may be required to enable MFA based on security policies or access to sensitive information.

BR-015 Session Security

Users shall be able to review active sessions, revoke sessions, and securely sign out from all devices.

Reporting

BR-016 Report Lifecycle

Every report shall follow the same lifecycle:

- Draft
- AI Processing
- Expert Review
- Approved
- Published
- Archived (optional)

Reports may only become visible to Clients after reaching the Published status.

BR-017 Version Control

Every published report shall retain version history.

Previous versions must remain available for authorized administrative users.

Projects

BR-018 Project Ownership

Every project must have:

- One owning Organization
- One responsible Admin
- One current status
- Developers may only access projects explicitly assigned to them.

BR-019 Approval Workflow

Project phases requiring formal approval shall not advance until the required approval has been recorded.

Audit & Compliance

BR-020 Audit Logging

All security-sensitive and business-critical actions shall be recorded in an immutable audit log.

Examples include:

- User authentication
- Role changes
- Report publication
- AI prompt changes
- Permission changes
- File uploads
- Invoice creation
- Project approvals
- Impersonation sessions

BR-021 Audit Visibility

Audit records may only be viewed by authorized users according to their assigned permissions.

Notifications

BR-022 Event-Driven Notifications

Notifications shall only be generated in response to defined business events.

Each notification must be linked to its originating event and recipient.

Billing

BR-023 Payment Confirmation

No paid service shall become active until payment has been successfully confirmed by the payment provider.

Technical Operations

BR-024 Technical Operator Access

Technical Operators shall receive configurable permissions defined by the Super Admin.

Permissions are not fixed and may be granted, modified, or revoked at any time.

Temporary access shall be supported.

BR-025 Technical Operator Accountability

Every action performed by a Technical Operator shall be logged with:

- User identity
- Timestamp
- Action performed

- Target resource
- Result
- Session information
- Future Business Rules

Additional business rules may be introduced in future platform versions.

New rules shall never invalidate or conflict with existing approved business rules without formal revision of this specification.

Automation

BR-026 Workflow Automation

The platform shall support automation of routine business workflows, including lead capture and routing, record creation, follow-up actions, notifications, and sales pipeline progression, when predefined business conditions are satisfied.

Automated actions shall comply with all other Business Rules. Any action that would otherwise require human authorization under **BR-008** or **BR-009** shall not be automated unless explicitly permitted by platform configuration.

Authorized users may override automated actions, and every automated action shall be recorded in the audit log in accordance with **BR-020**.

CHAPTER 5

5. Global User Roles & Permissions

5.1 Purpose

This chapter defines the global user roles, responsibilities, and access boundaries within the IO SKY Platform.

The objective is to ensure that every authenticated user receives only the permissions required to perform their responsibilities while maintaining platform security, organizational isolation, and operational integrity.

Authorization throughout the platform shall follow the Principle of Least Privilege as defined in **PP-001** and **BR-013**.

Permissions defined in this chapter apply across all modules unless explicitly overridden by a more restrictive module-specific rule.

5.2 Platform Roles

The IO SKY Platform includes the following user roles:

- Client
- Developer
- Admin
- Super Admin
- Technical Operator
- No additional platform roles are included in

Version 1 unless approved through a formal revision of this specification.

5.3 Client

Purpose

The Client represents the customer organization using the IO SKY Platform.

Clients access the platform to complete AI

Scans, review reports, monitor project progress, manage organizational documents, and communicate with IO SKY.

Clients never perform administrative or platform management tasks.

Permissions

A Client may:

- Access the Client Portal.
- Complete purchased AI Scans.
- View published reports.
- Download approved reports.
- View assigned projects.
- Track project progress.
- View invoices.
- Download invoices.
- Access uploaded project documentation.
- Send messages to Admins.
- Receive notifications.
- Manage personal profile settings.
- Configure personal MFA settings.
- Restrictions
- A Client shall not:
- Access another Organization.
- View unpublished reports.
- Modify reports.
- Access platform settings.
- Access CRM data.
- View other customer organizations.
- Communicate directly with Developers.
- Assign tasks.
- Modify project schedules.
- Publish documentation.

5.4 Developer

Purpose

Developers are responsible for delivering assigned technical work.

Developers operate exclusively within projects assigned to them.

Permissions

A Developer may:

- Access the Developer Portal.

- View assigned projects.
- View assigned tasks.
- Upload project files.
- Upload technical documentation.
- Update task status.
- Record working hours.
- Communicate internally with Admins and
- Super Admins.
- Receive internal notifications.
- Restrictions
- A Developer shall not:
- Access CRM.
- Access Billing.
- Access customer financial information.
- View customer AI Scan responses unless explicitly authorized.
- Modify user permissions.
- Publish reports.
- Access projects that are not assigned.

5.5 Admin

Purpose

Admins manage the daily operation of customer relationships, projects, AI Scans, communication, and delivery.

They serve as the primary operational contact between IO SKY and the customer.

Permissions

An Admin may:

- Manage customers.
- Manage projects.
- Manage AI Scans.
- Publish approved reports.
- Assign Developers.
- Manage tasks.
- Manage CRM records.
- Manage Discovery Calls.
- Manage project documentation.
- Manage customer communication.

- View billing information.
- Generate invoices where applicable.
- Manage notifications.
- Review dashboards and KPIs.
- Restrictions
- An Admin shall not:
- Create Super Admin accounts.
- Modify platform-wide security policies.
- Change platform configuration.
- Modify global integrations.
- Manage system infrastructure.

5.6 Super Admin

Purpose

The Super Admin is responsible for platform governance, security, configuration, compliance, and strategic administration.

The Super Admin manages the platform—not the day-to-day execution of customer projects.

Permissions

The Super Admin has unrestricted administrative access, including:

- Organization management.
- User management.
- Role management.
- Platform configuration.
- Global integrations.
- AI configuration.
- Prompt management.
- Audit review.
- Billing oversight.
- Analytics.
- System configuration.
- Impersonation.
- Security policies.
- Notification templates.
- Restrictions
- No functional restrictions apply within the platform.
- However, all Super Admin actions must be fully audited.

5.7 Technical Operator

Purpose

The Technical Operator is responsible for maintaining the technical health, stability, and operational continuity of the platform.

This role exists to perform infrastructure and technical maintenance without granting unrestricted administrative access.

Permissions are configurable and assigned by a

Super Admin.

Permissions

A Technical Operator may receive permission to:

- Monitor platform health.
- Review system logs.
- Monitor deployments.
- Execute deployments.
- Perform rollbacks.
- Manage infrastructure services.
- Manage storage.
- Review API health.
- Review queue processing.
- Monitor AI processing jobs.
- Restart platform services.
- Manage technical maintenance tasks.
- Access development environments.
- Perform database maintenance when explicitly authorized.
- Permissions shall be configurable on an individual basis.
- Restrictions
- Unless explicitly authorized, a Technical
- Operator shall not:
- View AI Scan responses.
- Read customer messages.
- View report contents.
- Access CRM.
- View invoices.
- Modify business data.

- Change user roles.
- Publish reports.

Access confidential customer information unrelated to technical maintenance.

5.8 Permission Matrix

Capability	Client	Developer	Admin	Super Admin	Technical Operator
Client Portal	✓	—	—	✓	—
Developer Portal	—	✓	—	✓	Optional
Admin Portal	—	—	✓	✓	Optional
Super Admin Portal	—	—	—	✓	—
Complete AI Scan	✓	—	—	—	—
View Published Reports	✓	Optional	✓	✓	No
Publish Reports	—	—	✓	✓	—
Manage Projects	View Only	Assigned Only	✓	✓	Optional
CRM Access	—	—	✓	✓	—
Billing	Own Only	—	✓	✓	—
User Management	—	—	Limited	✓	—
Platform Configuration	—	—	—	✓	—
Infrastructure Management	—	—	—	Optional	✓
Audit Logs	Own Activity	Own Activity	Limited	✓	Technical Logs Only
AI Prompt Management	—	—	—	✓	—

5.9 Authorization Model

All authorization decisions shall be based on:

- User Identity
- Assigned Role
- Assigned Permissions
- Organization Membership
- Resource Ownership

- Business Rules
- Platform Security Policies

No authorization decision shall rely solely on client-side validation.

5.10 Future Extensibility

The authorization model shall support additional roles, custom permission groups, and organization-specific permission profiles without requiring architectural redesign.

Developer Note

This chapter defines what each role is allowed to do. The implementation of authentication, authorization, Role-Based Access Control

(RBAC), permission storage, session management, and enforcement mechanisms is the responsibility of the development team.

CHAPTER 6

6. Global Platform Architecture

6.1 Purpose

This chapter defines the high-level architecture of the IO SKY Platform.

Its purpose is to describe how all functional modules interact as one integrated ecosystem while remaining independent enough to support future growth, scalability, and maintainability.

This chapter defines platform behaviour only.

The technical implementation, infrastructure, database architecture, and software design patterns remain the responsibility of the development team.

6.2 Platform Architecture Overview

The IO SKY Platform shall operate as a single integrated platform consisting of interconnected functional modules.

Each module is responsible for a specific business capability while securely exchanging information with other authorized modules through controlled platform services.

The platform shall not operate as separate applications.

All modules shall share a common authentication system, authorization model, notification framework, audit system, and organizational structure.

6.3 Core Platform Modules

Version 1 consists of the following core modules:

- Public Website & Lead Experience
- Identity & Authentication
- AI Scan Platform
- Client Portal
- Developer Portal
- Admin Portal
- Super Admin Portal
- CRM & Sales Management
- Project & Delivery Management
- Reporting

- Commercial Billing & Subscription
- Management
- Notifications
- File & Document Management
- AI Intelligence Layer
- Audit & Compliance
- Analytics & Business Intelligence
- Platform Configuration
- Integrations
- Platform Operations & Maintenance
- Global Non-Functional Requirements

Each module is specified independently but forms part of one unified platform.

6.4 Core Platform Services

The following services are considered platform- wide services.

They shall be shared by all modules where applicable.

Authentication Service

Responsible for:

- User authentication
- Session management
- Identity verification
- Multi-Factor Authentication
- Single Sign-On
- Authorization Service
- Responsible for:
- Role-Based Access Control (RBAC)
- Permission validation
- Organization isolation
- Access enforcement
- Notification Service
- Responsible for:
- Email notifications
- In-app notifications
- System alerts
- Business event notifications
- AI Service

- Responsible for:
- AI analysis
- AI report generation
- AI recommendations
- AI assistants
- AI Agents
- Reporting Service
- Responsible for:
- Report generation
- Report versioning
- Report publication
- Report storage
- Audit Service
- Responsible for:
- Audit logging
- Security events
- Compliance records
- Activity history
- File Management Service
- Responsible for:
- Secure file storage
- Version control
- Document sharing
- Download management
- Billing Service
- Responsible for:
- Payments
- Invoices
- Payment verification
- Subscription management

6.5 Module Communication

Principles

Modules shall communicate only through approved platform services.

Modules shall never bypass platform security controls.

Every data exchange shall respect:

- Authentication
- Authorization
- Organization ownership
- Audit logging
- Business Rules

6.6 Data Ownership

Every business entity shall have a single authoritative owner.

Examples include:

- Organization
- User
- AI Scan
- Report
- Project
- Invoice
- File
- Message
- Notification
- Task
- Discovery Call
- Duplicate business ownership is not permitted.

6.7 Event-Driven Platform

Behaviour

The platform shall operate using business events.

Examples include:

- AI Scan Purchased
- Payment Confirmed
- Report Approved
- Report Published
- Discovery Call Requested
- Project Created
- Task Assigned
- Invoice Paid
- User Invited

- Account Activated

Business events may trigger notifications, AI workflows, automation, CRM updates, or project creation.

6.8 Human Approval Workflow

The following activities require explicit human approval before completion:

- Report publication
- Customer-facing recommendations
- Proposal approval
- Project completion
- Invoice corrections
- User role changes with elevated privileges

AI-generated content shall never bypass mandatory human approval where defined by business rules.

6.9 Platform Integration Principles

External systems shall integrate through standardized platform integration services.

Version 1 includes support for integrations such as:

- Stripe
- Google
- Microsoft
- Email Provider
- Analytics Provider
- AI Provider
- IVR / Telephony Services

The development team is responsible for selecting the most appropriate implementation approach.

6.10 Scalability Principles

The platform shall be designed to support:

- Multiple Organizations
- Multiple Users per Organization
- Multiple Projects
- Multiple AI Scans
- Multiple Reports
- Multiple Admins
- Future functional modules without architectural redesign

The architecture shall support horizontal growth without requiring changes to the defined business behaviour.

6.11 Platform Design Principles

Every new feature introduced into the platform shall:

- Support operational efficiency.
- Improve user experience.
- Preserve platform consistency.
- Respect security requirements.
- Follow existing business rules.
- Integrate with the audit system where applicable.
- Support future scalability.

No feature shall introduce conflicting workflows or duplicate existing functionality.

6.12 Architectural Boundaries

This document intentionally does not define:

- Database schema
- SQL structure
- API implementation
- Framework selection
- Cloud infrastructure
- Programming language
- Design patterns
- Repository structure
- Deployment strategy

These implementation decisions remain the responsibility of the development team.

Developer Note

This chapter defines how the platform is organized from a business perspective.

It is intended to provide software architects and developers with a clear understanding of the platform's functional architecture while allowing full flexibility in the technical implementation.

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MODULE 1

7. Module 1 — Public Website & Lead Experience

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

7.1 Purpose

The Public Website serves as the primary entry point into the IO SKY Platform.

Its purpose is to educate visitors, establish trust, present IO SKY's services, qualify potential customers, and guide visitors toward meaningful business interactions.

The website is not intended to function as a marketing brochure. It is designed to operate as the beginning of the customer's operational journey within the IO SKY ecosystem.

7.2 Objectives

The Public Website shall enable visitors to:

- Understand IO SKY's positioning.
- Learn about available solutions and services.
- Request a Discovery Call.
- Contact IO SKY.
- Purchase an AI Scan.
- Access their account through secure authentication.
- Explore IO SKY’s custom architecture capabilities and implementation approach.
- Learn about the company.
- Submit partnership inquiries.
- Begin their customer journey.

7.3 Scope

The Public Website includes:

- Home
- Foundations
- Intelligence
- Solutions

- AI Scan
- Custom Architecture
- About
- Contact
- Discovery Call
- Login
- Legal Pages (Privacy Policy, Terms &
- Conditions, Cookie Policy)

No authenticated portal functionality is included within the Public Website.

7.4 Actors

The following actors interact with this module:

- Anonymous Visitor
- Prospective Customer
- Existing Client
- Admin
- Super Admin

7.5 Website Navigation

The primary navigation shall include:

- Home
- Foundations
- Intelligence
- Solutions
- AI Scan
- Custom Architecture
- About
- Contact
- Discovery Call
- Login

Navigation shall remain consistent across all public pages.

7.6 User Journey

Visitor Journey

Visitor arrives on the website.

Visitor explores services and solutions.

Visitor gains trust through company information and custom architecture examples.

Visitor chooses one of the available conversion paths.

Visitor submits information.

Platform validates submission.

Confirmation is displayed.

CRM record is created where applicable.

Appropriate notifications are generated.

Visitor continues into the IO SKY customer journey.

7.7 Primary Conversion Paths

The Public Website shall support the following conversion paths:

- Discovery Call
- Visitor submits a Discovery Call request.
- The platform creates:
- Lead record
- CRM activity
- Internal notification
- Confirmation email
- Contact Request
- Visitor submits a general inquiry.
- The platform creates:
- Contact record
- Internal notification
- Confirmation email
- AI Scan Purchase
- Visitor purchases an AI Scan.
- The platform begins the AI Scan onboarding workflow.
- Login

Existing users authenticate and are redirected automatically to their assigned portal.

7.8 Functional Requirements

The website shall:

- Present company positioning.
- Display service information.
- Display operational ecosystems.
- Present AI Scan offerings.
- Display custom architecture examples.
- Provide contact forms.
- Provide Discovery Call forms.
- Provide secure login.
- Display legal documentation.
- Support responsive design.
- Support accessibility standards.
- Support multilingual expansion in future versions.

7.9 Business Rules

Rule	Business Rule
BR-001	Authentication Required
BR-003	Automatic Portal Routing
BR-004	Invitation-Based Access
BR-022	Event-Driven Notifications

No additional module-specific business rules are defined.

7.10 Integrations

The Public Website integrates with:

- CRM
- Authentication Service
- Billing Service
- Notification Service
- AI Scan Platform
- Analytics Service

7.11 Notifications

The platform shall generate notifications for:

- Discovery Call submitted
- Contact request submitted
- AI Scan purchased

- Partnership request submitted
- Form submission failed (where applicable)

Notifications shall be delivered to the appropriate administrative users.

7.12 Security Requirements

The Public Website shall:

- Validate all submitted data.
- Protect against automated spam submissions.
- Prevent unauthorized access to protected resources.
- Enforce HTTPS for all communications.
- Never expose internal platform information.
- Log security-related events where applicable.

7.13 Acceptance Criteria

The module shall be considered complete when:

- All public pages are accessible.
- Navigation behaves consistently.
- All forms validate correctly.
- CRM records are created successfully.
- Notifications are generated correctly.
- Login redirects users to the correct portal.
- AI Scan purchases initiate the onboarding workflow.
- The website is fully responsive.
- Legal pages are accessible.

7.14 Out of Scope

The following functionality is excluded from

Version 1 of this module:

- Public user registration.
- Customer dashboards.
- Project management.
- AI report viewing.
- Internal administration.
- Developer functionality.
- Billing management.
- CRM management.

- These functions are provided by authenticated platform modules.

7.15 Developer Notes

This module defines the required business behaviour of the Public Website.

The development team is responsible for determining the implementation details, including frontend framework, routing architecture, content management approach, performance optimization, and technical integrations, provided that all functional requirements defined in this specification are satisfied.

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MODULE 2

8. Module 2 — Identity & Authentication

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

8.1 Purpose

The Identity & Authentication module is responsible for securely identifying users, authenticating access, enforcing authorization policies, and protecting all restricted areas of the

IO SKY Platform.

This module provides a unified identity system across all platform modules and ensures that every authenticated user is automatically directed to the correct portal based on their assigned role and permissions.

8.2 Objectives

The module shall enable the platform to:

- Authenticate users securely.
- Authorize access based on assigned roles and permissions.
- Protect platform resources.
- Manage user identities.
- Support secure account activation.
- Manage password recovery.
- Support Multi-Factor Authentication (MFA).
- Manage user sessions.
- Automatically route users to the correct portal.
- Record authentication-related audit events.

8.3 Scope

The module includes:

- User Authentication
- Identity Verification
- Account Activation

- Password Management
- Password Reset
- Session Management
- Multi-Factor Authentication
- Role-Based Access Control
- Portal Routing
- Account Invitations
- Audit Logging for Authentication Events

User profile management is handled within the relevant authenticated portals.

8.4 Actors

The following actors interact with this module:

- Client
- Developer
- Admin
- Super Admin
- Technical Operator

Anonymous visitors may only access the login page and password recovery functionality.

8.5 Authentication Principles

The authentication system shall operate according to the following principles:

- Every user has a unique identity.
- Authentication occurs once per session.
- Authorization is evaluated on every protected request.

Users are automatically redirected to their assigned portal after successful authentication.

Authentication status alone does not determine access; authorization policies must also be satisfied.

8.6 Account Creation

Public self-registration is not permitted.

Accounts may only be created through approved platform workflows:

- AI Scan purchase and onboarding.
- Invitation by an Admin.
- Invitation by a Super Admin.
- Internal onboarding of Developers.
- Internal onboarding of Technical Operators.

- Each new account must be associated with an

Organization before activation, except for platform-level users such as Super Admins.

8.7 Account Activation

Newly created accounts shall remain inactive until the user completes the activation process.

The activation workflow shall include:

- Account invitation or creation.
- Activation email sent.
- Identity verification.
- Password creation.
- Acceptance of applicable legal agreements.
- Optional MFA enrollment, depending on role.
- Account activation.
- Automatic sign-in.
- Automatic portal routing.

An account may not access protected resources until activation has been successfully completed.

8.8 Authentication Workflow

The standard authentication process shall follow these steps:

- User enters credentials.
- Credentials are validated.
- MFA challenge is presented if required.
- Authentication succeeds.
- Authorization policies are evaluated.
- User role is resolved.
- Organization membership is verified.
- Active permissions are loaded.
- Audit event is recorded.
- User is redirected to the appropriate portal.

8.9 Password Management

The platform shall support:

- Secure password creation.
- Password updates.
- Password recovery.

- Password reset via verified email.

Password expiration policies where required by platform configuration.

Passwords shall never be recoverable in plain text.

8.10 Multi-Factor Authentication

MFA is mandatory for:

- Super Admin
- Admin
- Technical Operator
- Developers are strongly encouraged to enable
- MFA.

Clients may be required to enable MFA based on organizational policy or access to sensitive information.

The platform shall support future expansion to additional authentication methods.

8.11 Session Management

The platform shall:

- Create secure authenticated sessions.
- Allow users to view active sessions.
- Allow users to terminate individual sessions.
- Allow users to sign out from all devices.

Automatically expire inactive sessions according to configurable platform policies.

8.12 Portal Routing

Following successful authentication, users shall be redirected automatically according to their assigned role.

Manual portal selection shall not be required.

If a user has multiple authorized roles, the routing behaviour shall follow platform rules defined by the Super Admin.

8.13 Authorization

Authorization decisions shall consider:

- User identity.
- Assigned role.
- Explicit permissions.

- Organization membership.
- Ownership of requested resources.
- Applicable Business Rules.
- Platform security policies.
- Access shall be denied by default unless explicitly permitted.

8.14 Audit Requirements

The platform shall record authentication-related events, including:

- Successful sign-in.
- Failed sign-in.
- Account activation.
- Password changes.
- Password reset requests.
- MFA enrollment.
- MFA removal.
- Session termination.
- Account lockout.
- Permission changes affecting authentication.

8.15 Business Rules

Rule	Business Rule
BR-001	Authentication Required
BR-002	Single Identity
BR-003	Automatic Portal Routing
BR-004	Invitation-Based Access
BR-013	Least Privilege
BR-014	Multi-Factor Authentication
BR-015	Session Security

8.16 Acceptance Criteria

The module shall be considered complete when:

- Only authorized users can access protected resources.
- Public registration is disabled.
- Account activation functions correctly.

- Password recovery operates securely.
- MFA functions as specified.
- Session management is operational.
- Users are automatically routed to the correct portal.
- Authentication events are recorded in the audit log.
- Unauthorized access attempts are denied and logged.

8.17 Out of Scope

The following functionality is excluded from this module:

- User profile editing.
- Organization management.
- Role administration.
- Billing.
- CRM.
- Project management.
- AI Scan processing.
- These functions are provided by other modules.

8.18 Developer Notes

This module defines the required business behaviour for identity and authentication.

The development team is responsible for selecting and implementing the appropriate authentication technologies, identity providers, session management mechanisms, and authorization architecture, provided that all functional and security requirements defined in this specification are satisfied.

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MODULE 3

9. Module 3 — AI Scan Platform

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

9.1 Purpose

The AI Scan Platform enables organizations to assess their operational maturity, identify automation opportunities, evaluate cybersecurity risks, and receive AI-assisted business insights.

The AI Scan serves as the primary entry point into the IO SKY service ecosystem and is designed to generate qualified business opportunities while delivering measurable value to customers.

The AI Scan Platform combines structured data collection, artificial intelligence, expert validation, and customer reporting into one standardized workflow.

9.2 Objectives

The AI Scan Platform shall enable organizations to:

- Purchase an AI Scan.
- Complete a structured assessment questionnaire.
- Submit business information securely.
- Receive AI-assisted analysis.
- Receive expert-reviewed reports.
- Access approved reports through the Client
- Portal.
- Track scan progress.
- Request follow-up services.
- Book a Discovery Call after report publication.

9.3 Scope

The module includes:

- AI Scan product selection
- Secure payment
- Account creation

- Questionnaire
- Progress saving
- AI analysis
- Expert review
- Report approval
- Report publication
- Report versioning
- Customer notifications
- Client Portal integration
- Audit logging

9.4 Actors

The following actors interact with this module:

- Prospective Customer
- Client
- Admin
- Super Admin
- AI Intelligence Layer

9.5 AI Scan Lifecycle

Every AI Scan shall follow the same mandatory lifecycle.

No scan may bypass any required stage.

Stage 1 – Purchase

The customer selects an AI Scan package and completes payment.

Payment must be successfully confirmed before the workflow continues.

Stage 2 – Account Creation

If the customer does not already have an account, the platform creates one.

The customer completes the account activation process as defined in the Identity &

Authentication module.

Stage 3 – Questionnaire

The customer completes the AI Scan questionnaire.

The platform shall allow:

- Draft saving
- Resume later
- Validation of required fields
- Secure submission
- Stage 4 – AI Processing

After submission, the AI Intelligence Layer analyzes the provided information.

The AI may generate:

- Operational insights
- Automation opportunities
- Risk observations
- Business recommendations
- Executive summary
- Supporting analysis
- The generated report remains an internal draft.
- Stage 5 – Expert Review
- An authorized IO SKY reviewer evaluates the AI- generated report.
- The reviewer may:
- Approve the report.
- Request modifications.
- Reject the draft.
- Regenerate specific sections.
- Add manual recommendations.
- Edit customer-facing content.
- The report remains unavailable to the customer during this stage.
- Stage 6 – Publication

Only after explicit approval by an authorized reviewer may the report be published.

Publication shall:

- Make the report visible in the Client Portal.
- Trigger customer notifications.
- Record audit events.
- Create the first published version.
- Stage 7 – Customer Access
- After publication the customer may:
- View the report.
- Download the report.
- Review recommendations.

- Request a Discovery Call.
- Continue into implementation services.

9.6 Report Status Model

Each report shall always have exactly one status:

- Draft
- Questionnaire In Progress
- Submitted
- AI Processing
- Awaiting Expert Review
- Revision Required
- Approved
- Published
- Archived
- Transitions between statuses shall follow the approved workflow.

9.7 Functional Requirements

The AI Scan Platform shall:

- Support multiple scan packages.
- Process secure payments.
- Create customer accounts when required.
- Support resumable questionnaires.
- Validate required responses.
- Generate AI-assisted reports.
- Require expert approval before publication.
- Maintain report version history.
- Publish approved reports to the Client Portal.
- Support future scan templates.

9.8 Business Rules

Rule	Business Rule
BR-004	Invitation-Based Access
BR-008	AI Supports Human Decision-Making
BR-009	Human Approval Required
BR-010	AI Transparency

BR-016	Report Lifecycle
BR-017	Version Control
BR-020	Audit Logging

No report shall become visible until it reaches the Published status.

9.9 Notifications

The platform shall generate notifications for:

- Payment confirmed
- Account activation required
- Questionnaire available
- Questionnaire submitted
- AI processing started
- Expert review started
- Revision requested
- Report approved
- Report published
- Discovery Call requested

Notification recipients shall be determined by platform rules and user roles.

9.10 Security Requirements

The AI Scan Platform shall:

- Protect all submitted customer information.
- Restrict report access to authorized users only.
- Prevent unauthorized report publication.
- Record all approval actions.
- Log all report downloads.
- Validate customer ownership before granting report access.

9.11 Acceptance Criteria

The module shall be considered complete when:

- Customers can successfully purchase an AI
- Scan.
- Accounts are created and activated correctly.
- Questionnaires can be completed and resumed.
- AI analysis is generated successfully.

- Expert review is mandatory before publication.
- Reports are published only after approval.
- Published reports are accessible through the
- Client Portal.
- All workflow stages are recorded in the audit log.
- Notifications are delivered at each defined milestone.

9.12 Out of Scope

The following functionality is excluded from

Version 1:

- Fully autonomous report publication.
- AI-generated invoices.
- AI-generated contracts.
- Automated implementation planning without human review.
- Third-party report sharing without authorization.

9.13 Developer Notes

This module defines the complete business workflow for the AI Scan Platform.

The development team is responsible for implementing the workflow, AI integrations, report generation, storage, payment processing, and publication mechanisms while ensuring strict compliance with the business rules and approval process defined in this specification.

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MODULE 4

10. Module 4 — Client Portal

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

10.1 Purpose

The Client Portal serves as the secure operational workspace for customer organizations within the IO SKY Platform.

It provides authorized users with centralized access to their AI Scan reports, projects, documents, invoices, notifications, and communication with IO SKY.

The Client Portal is designed to provide transparency, visibility, and control throughout the customer's journey while ensuring that each organization can access only its own information.

10.2 Objectives

The Client Portal shall enable authorized clients to:

- Access their personalized dashboard.
- View AI Scan progress and results.
- Access published reports.
- Monitor project progress.
- Download project documentation.
- Upload documents
- View invoices and payment status.
- Receive notifications.
- Communicate with IO SKY.
- Manage personal account settings.
- Request additional services.

10.3 Scope

The Client Portal includes:

- Dashboard
- AI Scan Center

- Reports
- Projects
- Documents
- Invoices
- Messages
- Notifications
- Profile & Security
- Service Requests
- Organization management is outside the scope of this module.

10.4 Actors

The following actors interact with this module:

- Client
- Admin
- Super Admin
- Developers do not have direct access to the

Client Portal unless explicitly authorized through an administrative impersonation or support workflow.

10.5 Dashboard

After authentication, the Client shall arrive at a personalized dashboard displaying relevant information for their organization.

The dashboard shall provide an overview of:

- Active AI Scans
- Recently published reports
- Active projects
- Upcoming milestones
- Recent notifications
- Outstanding invoices
- Recent messages
- Quick actions

The dashboard shall present only information that the authenticated user is authorized to access.

10.6 AI Scan Center

The AI Scan Center shall allow clients to:

- View purchased AI Scans.

- View current scan status.
- Resume incomplete questionnaires.
- Track review progress.
- Access published reports.
- View report history.
- Request follow-up consultations.

Reports that have not reached the Published status shall not be accessible.

10.7 Reports

Clients shall be able to:

- View published reports.
- Download reports.
- View report versions where permitted.
- Review executive summaries.
- Review recommendations.
- View publication dates.
- View approval history where applicable.
- Clients shall not modify report content.

10.8 Project Management

Clients shall be able to:

- View assigned projects.
- Review project status.
- View milestones.
- View completed deliverables.
- View upcoming activities.
- Review project documentation.
- Monitor implementation progress.

Clients shall not modify internal project planning unless explicitly permitted.

10.9 Document Center

The Document Center shall provide access to:

- Published reports.
- Project documentation.
- Signed agreements.
- Technical documentation intended for customers.

- User guides.
- Downloadable resources.
- All documents shall remain associated with the owning Organization.

10.10 Billing

Clients shall be able to:

- View invoices.
- View payment status.
- Download invoices.
- View payment history.
- Access billing-related documents.
- Clients shall not modify billing records.

10.11 Messages

The Client Portal shall include secure communication with IO SKY.

Clients may:

- Send messages to authorized Admins.
- Receive replies.
- View conversation history.
- Upload supporting files where permitted.

Direct communication with Developers is prohibited in accordance with **BR-011**.

10.12 Notifications

Clients shall receive notifications regarding:

- AI Scan updates.
- Report publication.
- Project milestones.
- Invoice availability.
- Payment confirmations.
- New messages.
- Service announcements.
- Security-related events.

Notification preferences shall be configurable by the user where permitted.

10.13 Profile & Security

Clients shall be able to:

- Update personal profile information.
- Change password.
- Configure Multi-Factor Authentication where supported.
- Review active sessions.
- Terminate active sessions.
- Review account activity.
- Manage notification preferences.

Changes affecting Organization ownership or permissions are outside the scope of this module.

10.14 Service Requests

The Client Portal shall allow customers to submit requests for additional services, including:

- Discovery Call requests.
- Support requests.
- Project inquiries.
- Additional AI Scans.
- Consultation requests.

Each request shall generate appropriate CRM records, notifications, and audit events.

10.15 Business Rules

Rule	Business Rule
BR-001	Authentication Required
BR-006	Organization Isolation
BR-007	Data Ownership
BR-011	Customer Communication
BR-016	Report Lifecycle
BR-017	Version Control
BR-020	Audit Logging

10.16 Security Requirements

The Client Portal shall:

- Restrict access to authenticated users only.
- Verify Organization ownership for every protected resource.
- Prevent access to unpublished reports.
- Protect sensitive customer information.
- Record access to business-critical documents.
- Log downloads of published reports.

10.17 Acceptance Criteria

The Client Portal shall be considered complete when:

Clients are automatically redirected to the portal after authentication.

Dashboards display only authorized information.

Published reports are accessible.

AI Scan progress is visible.

Project information is displayed correctly.

Documents can be downloaded securely.

Invoices are available.

Messaging functions correctly.

Notifications are delivered successfully.

All activities are recorded in the audit log where required.

10.18 Out of Scope

The following functionality is excluded from

Version 1:

- Editing AI-generated reports.
- Direct Developer communication.
- Internal project planning.
- User role administration.
- Organization administration.
- Platform configuration.
- CRM management.

10.19 Developer Notes

This module defines the business behaviour of the Client Portal.

The development team is responsible for implementing the portal's user interface, navigation, document management, messaging capabilities, billing integration, and security controls while ensuring compliance with the business rules defined throughout this specification.

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MODULE 5

11. Module 5 — Developer Portal

Version	1.1	**Priority**	High
Status	Release Candidate	**Business Owner**	IO SKY

11.1 Purpose

The Developer Portal provides a secure workspace for Developers to access assigned projects, manage technical tasks, upload deliverables, record progress, and collaborate internally with IO SKY.

The Developer Portal is strictly an internal operational environment. It is not intended for customer interaction and shall never expose confidential customer information beyond what is required to complete assigned work.

11.2 Objectives

The Developer Portal shall enable Developers to:

- View assigned projects.
- Manage assigned tasks.
- Upload technical deliverables.
- Update development progress.
- Record working hours.
- Access technical documentation.
- Communicate internally with authorized IO
- SKY personnel.
- Receive project-related notifications.

11.3 Scope

The Developer Portal includes:

- Developer Dashboard
- Assigned Projects
- Task Management
- File & Deliverable Management
- Time Registration

- Internal Messaging
- Notifications
- Technical Documentation
- Profile & Security

Project administration, customer management, billing, and CRM functionality are outside the scope of this module.

11.4 Actors

The following actors interact with this module:

- Developer
- Admin
- Super Admin
- Technical Operator (where authorized)
- Clients shall never access the Developer Portal.

11.5 Developer Dashboard

Upon authentication, Developers shall be presented with a dashboard displaying information relevant to their assigned work.

The dashboard shall include:

- Assigned projects.
- Open tasks.
- Upcoming deadlines.
- Recently assigned work.
- Pending file reviews.
- Recent internal messages.
- Notifications.
- Time registration summary.

The dashboard shall display only data that the authenticated Developer is authorized to access.

11.6 Project Workspace

Developers shall be able to access only projects explicitly assigned to them.

For each assigned project, the Developer may:

- View project information.
- Review project objectives.
- Access approved technical documentation.

- View assigned milestones.
- Access assigned deliverables.
- Review project updates.
- View project deadlines.

Developers shall not modify project ownership, budgets, customer information, or commercial data.

11.7 Task Management

The platform shall provide structured task management.

Developers may:

- View assigned tasks.
- Update task status.
- Record progress notes.
- Mark tasks as completed.
- Request clarification from an Admin.
- Attach technical comments.

Task assignment remains the responsibility of authorized administrative users.

11.8 Deliverable Management

Developers shall be able to:

- Upload source deliverables.
- Upload technical documentation.
- Upload implementation notes.
- Replace previous versions where permitted.
- View upload history.

Published customer documentation shall only become available after administrative approval.

11.9 Time Registration

The platform shall support recording development time.

Developers may:

- Log working hours.
- Associate time entries with projects and tasks.
- Add optional work descriptions.
- Review historical time entries.
- Time records may be reviewed by authorized
- Admins and Super Admins.

11.10 Internal Communication

The Developer Portal shall support secure internal communication.

Developers may communicate with:

- Assigned Admins.
- Super Admins.
- Authorized Technical Operators.
- Developers shall not communicate directly with
- Clients through the platform.

11.11 Notifications

Developers shall receive notifications for:

- New project assignments.
- New task assignments.
- Task updates.
- Project changes.
- Internal messages.
- Documentation requests.
- Deadline reminders.
- Administrative announcements.

11.12 Profile & Security

Developers shall be able to:

- Update personal profile information.
- Change password.
- Configure Multi-Factor Authentication where supported.
- Review active sessions.
- Terminate active sessions.
- Manage notification preferences.

Developers shall not modify platform permissions or organizational settings.

11.13 Business Rules

Rule	Business Rule
BR-001	Authentication Required
BR-006	Organization Isolation

BR-011	Customer Communication
BR-013	Least Privilege
BR-014	Multi-Factor Authentication
BR-020	Audit Logging

11.14 Security Requirements

The Developer Portal shall:

- Restrict access to assigned projects only.

Prevent access to commercial customer information unless explicitly authorized.

Protect uploaded technical files.

Record all file uploads and modifications.

Log project access and task updates.

Verify permissions before granting access to project resources.

11.15 Acceptance Criteria

The module shall be considered complete when:

Developers are automatically redirected to the Developer Portal after authentication.

Only assigned projects are visible.

Task management functions correctly.

Deliverables can be uploaded securely.

Time registration operates correctly.

Internal messaging functions as specified.

Notifications are delivered successfully.

All business-critical actions are recorded in the audit log.

11.16 Out of Scope

The following functionality is excluded from

Version 1:

- Customer communication.

- CRM management.
- Billing management.
- User administration.
- Platform configuration.
- AI report publication.
- Project approval workflows.
- These capabilities belong to other platform modules.

11.17 Developer Notes

This module defines the functional requirements for the Developer Portal from a Product Owner perspective.

The development team is responsible for implementing task management, document handling, time registration, and internal collaboration features while ensuring compliance with the platform-wide business rules, security requirements, and authorization model.

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MODULE 6

12. Module 6 — Admin Portal

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

12.1 Purpose

The Admin Portal serves as the primary operational workspace for authorized IO SKY personnel.

It enables Admins to manage customers, AI

Scans, reports, projects, Developers, communication, billing, and operational workflows from a centralized environment.

The Admin Portal provides operational visibility and control while ensuring that all actions comply with platform-wide security, audit, and business rules.

12.2 Objectives

The Admin Portal shall enable authorized

Admins to:

- Manage customer organizations.
- Review and manage AI Scans.
- Review and publish approved reports.
- Manage projects.
- Assign Developers.
- Manage customer communication.
- Monitor operational progress.
- Manage billing activities.
- Track platform notifications.
- View operational dashboards.

12.3 Scope

The Admin Portal includes:

- Operational Dashboard

- Customer Management
- AI Scan Management
- Report Management
- Project Management
- Developer Assignment
- Communication Center
- Billing Overview
- Notifications
- Operational Analytics

Platform-wide configuration and infrastructure management are outside the scope of this module.

12.4 Actors

The following actors interact with this module:

- Admin
- Super Admin
- Technical Operator (where explicitly authorized)

Clients and Developers shall not have direct access to the Admin Portal.

12.5 Operational Dashboard

Upon authentication, Admins shall be presented with a dashboard displaying operational metrics and work requiring attention.

The dashboard shall include:

- Active customer organizations.
- AI Scans awaiting review.
- Reports awaiting publication.
- Active projects.
- Developer workload overview.
- Upcoming project milestones.
- Outstanding invoices.
- Recent customer communication.
- Platform notifications.
- Operational KPIs.

12.6 Customer Management

Admins shall be able to:

- View customer organizations.
- Create customer organizations.
- Update customer information.
- Manage customer contacts.
- Review customer history.
- View purchased services.
- View AI Scan history.
- View project portfolio.
- View customer documents.
- Customer deletion is outside the scope of this module.

12.7 AI Scan Management

Admins shall be able to:

- Monitor AI Scan progress.
- Review submitted questionnaires.
- Track AI processing.
- Assign reports for expert review.
- Request additional customer information.
- Return questionnaires for correction where applicable.
- View scan history.

Admins shall not bypass mandatory review stages defined in the Business Rules.

12.8 Report Management

Admins shall be able to:

- Review AI-generated reports.
- Edit customer-facing content.
- Add manual recommendations.
- Approve reports.
- Reject reports.
- Publish approved reports.
- Archive reports.
- View report version history.
- Every publication action shall be recorded in the audit log.

12.9 Project Management

Admins shall be able to:

- Create projects.
- Assign Developers.
- Update project status.
- Define milestones.
- Upload project documentation.
- Review project progress.
- Close completed projects.
- Monitor project risks.

12.10 Developer Management

Admins shall be able to:

- View available Developers.
- Assign Developers to projects.
- Review Developer workload.
- Review task completion.
- Review recorded working hours.
- Reassign work where necessary.
- Admins shall not modify Developer platform permissions.

12.11 Communication Center

The Admin Portal shall provide centralized communication with customers.

Admins shall be able to:

- View customer conversations.
- Reply to customer messages.
- Send announcements.
- Manage communication history.
- Upload attachments where applicable.

Direct customer communication with Developers shall remain prohibited.

12.12 Billing Overview

Admins shall be able to:

- View invoices.
- Review payment status.
- Generate invoices where authorized.
- View transaction history.
- Monitor outstanding payments.

Modification of payment provider settings is outside the scope of this module.

12.13 Notifications

Admins shall receive notifications for:

- New AI Scan purchases.
- Questionnaire submissions.
- Reports awaiting review.
- Project updates.
- Developer assignments.
- Customer messages.
- Payment confirmations.
- System alerts.
- Security events relevant to their responsibilities.

12.14 Operational Analytics

The Admin Portal shall provide operational insights including:

- Active Organizations.
- AI Scan conversion rates.
- Report publication statistics.
- Project completion metrics.
- Developer utilization.
- Customer activity.
- Service adoption.
- Operational workload.
- The analytics presented shall support operational decision-making.

12.15 Business Rules

Rule	Business Rule
BR-001	Authentication Required
BR-006	Organization Isolation
BR-009	Human Approval Required
BR-013	Least Privilege
BR-016	Report Lifecycle
BR-017	Version Control

BR-018	Project Ownership
BR-019	Approval Workflow
BR-020	Audit Logging

12.16 Security Requirements

The Admin Portal shall:

- Restrict access to authorized Admins only.
- Record all business-critical administrative actions.
- Prevent unauthorized report publication.
- Protect customer information.
- Verify permissions before administrative actions.
- Log approvals, assignments, and publication events.

12.17 Acceptance Criteria

The module shall be considered complete when:

- Admins are automatically redirected to the
- Admin Portal after authentication.
- Customer management functions correctly.
- AI Scans can be reviewed.
- Reports require explicit approval before publication.
- Projects can be managed successfully.
- Developers can be assigned to projects.
- Billing information is available.
- Operational dashboards display accurate information.
- Notifications function correctly.
- All required audit events are recorded.

12.18 Out of Scope

The following functionality is excluded from

Version 1:

- Platform infrastructure management.
- Global platform configuration.
- Authentication provider configuration.
- Organization-wide security policy management.
- Super Admin functions.

- These capabilities belong to the Super Admin
- Portal.

12.19 Developer Notes

This module defines the operational requirements for the Admin Portal.

The development team is responsible for implementing the administrative workflows, dashboards, review interfaces, project management features, and operational controls while ensuring compliance with all platform- wide business rules, security requirements, and audit obligations.

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MODULE 7

13. Module 7 — Super Admin Portal

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

13.1 Purpose

The Super Admin Portal provides centralized governance, platform administration, security oversight, and strategic configuration for the entire IO SKY Platform.

Unlike the Admin Portal, which focuses on day- to-day operations, the Super Admin Portal is responsible for platform-wide management, security, compliance, and system governance.

All actions performed within this module have platform-wide impact and therefore require comprehensive audit logging.

13.2 Objectives

The Super Admin Portal shall enable authorized

Super Admins to:

- Manage organizations.
- Manage platform users.
- Configure platform-wide settings.
- Manage global permissions.
- Configure AI behaviour.
- Monitor platform health.
- Review audit logs.
- Configure integrations.
- Manage notification templates.
- Monitor business intelligence dashboards.
- Configure security policies.

13.3 Scope

The Super Admin Portal includes:

- Executive Dashboard
- Organization Management
- User Management
- Role & Permission Management
- AI Configuration
- Security Center
- Audit Center
- Platform Configuration
- Integration Management
- Notification Management
- Business Intelligence
- Technical Operator Management
- Daily customer operations are outside the scope of this module.

13.4 Actors

The following actors interact with this module:

- Super Admin

Technical Operators may access only the specific platform functions explicitly assigned to them by a Super Admin.

13.5 Executive Dashboard

The Executive Dashboard shall provide a real- time overview of the entire platform.

The dashboard shall include:

- Total organizations.
- Active users.
- Active AI Scans.
- Reports awaiting review.
- Platform health.
- Security alerts.
- Infrastructure status.
- Revenue overview.
- Operational KPIs.
- Recent critical events.

13.6 Organization Management

Super Admins shall be able to:

- Create organizations.
- Update organizations.
- Suspend organizations.
- Reactivate organizations.
- View organization history.
- Review service subscriptions.
- Review organizational activity.

Deletion of organizations shall follow a controlled administrative workflow.

13.7 User & Role Management

Super Admins shall be able to:

- Create users.
- Invite users.
- Disable users.
- Reactivate users.
- Reset user access.
- Assign roles.
- Modify permissions.
- Review login history.
- Review account activity.
- Every permission change shall be recorded in the audit log.

13.8 Technical Operator

Management

Super Admins shall define the permissions assigned to each Technical Operator.

Permissions shall be individually configurable and may include:

- Infrastructure monitoring.
- Deployment management.
- System maintenance.
- Database maintenance.
- Queue management.
- Storage management.
- Log access.
- Performance monitoring.

Temporary permissions shall be supported and may be revoked at any time.

13.9 AI Configuration

Super Admins shall be able to configure platform-wide AI behaviour, including:

- AI model selection.
- Prompt configuration.
- AI feature availability.
- Report generation settings.
- AI processing limits.
- AI safety controls.
- Changes to AI configuration shall be versioned and fully auditable.

13.10 Security Center

The Security Center shall enable Super Admins to:

- Review security alerts.
- Monitor authentication activity.
- Review failed login attempts.
- Monitor Multi-Factor Authentication adoption.
- Review active sessions.
- Review suspicious activity.
- Configure platform security policies.

13.11 Audit Center

The Audit Center shall provide centralized access to immutable audit records.

Super Admins shall be able to:

- Search audit events.
- Filter audit records.
- Review administrative actions.
- Review authentication events.
- Review report publication history.
- Review permission changes.
- Export audit records where permitted.
- Audit records shall not be editable.

13.12 Platform Configuration

Super Admins shall be able to configure platform-wide settings including:

- Global business settings.

- Organization defaults.
- Notification defaults.
- Feature availability.
- Maintenance mode.
- Localization settings.
- Configuration changes shall be validated before becoming active.

13.13 Integration Management

Super Admins shall be able to manage platform integrations.

Supported integrations may include:

- Payment providers.
- AI providers.
- Email providers.
- Authentication providers.
- Analytics providers.
- Calendar providers.
- Communication services.

The implementation of each integration is the responsibility of the development team.

13.14 Notification Management

The Super Admin Portal shall support management of:

- Email templates.
- In-app notification templates.
- System notifications.
- Business event notifications.
- Changes shall apply platform-wide.

13.15 Business Intelligence

The Super Admin Portal shall provide executive- level insights, including:

- Platform growth.
- Customer acquisition.
- AI Scan conversion.
- Revenue trends.
- Project completion rates.
- User activity.
- Platform usage.

- Operational performance.

These dashboards are intended for strategic decision-making rather than daily operational management.

13.16 Business Rules

Rule	Business Rule
BR-001	Authentication Required
BR-006	Organization Isolation
BR-013	Least Privilege
BR-014	Multi-Factor Authentication
BR-020	Audit Logging
BR-021	Audit Visibility
BR-024	Technical Operator Access

13.17 Security Requirements

The Super Admin Portal shall:

- Require Multi-Factor Authentication.
- Record all administrative actions.
- Protect platform-wide configuration.
- Restrict access to authorized Super Admins only.
- Validate permission changes before application.
- Maintain complete audit trails.

13.18 Acceptance Criteria

The module shall be considered complete when:

- Super Admins can manage organizations.
- Users and permissions can be managed securely.
- Technical Operator permissions are configurable.
- AI configuration is versioned and auditable.
- Security monitoring functions correctly.
- Audit logs are searchable.
- Platform configuration changes are validated.
- Integration settings can be managed.

- Executive dashboards display accurate information.
- All platform-wide actions are recorded in the audit log.

13.19 Out of Scope

The following functionality is excluded from

Version 1:

- Direct software development.
- Customer project execution.
- Customer communication workflows.
- Developer task management.
- These capabilities belong to other platform modules.

13.20 Developer Notes

This module defines the governance and platform administration requirements for the

Super Admin Portal.

The development team is responsible for implementing secure administrative interfaces, configuration management, audit functionality, and platform governance mechanisms while ensuring compliance with the security, authorization, and audit requirements defined throughout this specification.

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MODULE 8

14. Module 8 — CRM & Sales Management

Version	1.1	Priority	Critical
Status	Release Candidate	Business Owner	IO SKY

14.1 Purpose

The CRM & Sales Management module provides a centralized environment for managing the complete customer acquisition lifecycle, from the first interaction through project handover.

The module enables IO SKY to capture, qualify, manage, and convert leads while maintaining a complete history of all commercial activities.

Every lead, customer interaction, proposal, and sales opportunity shall be managed within a structured and auditable workflow.

14.2 Objectives

The CRM & Sales Management module shall enable authorized users to:

- Capture leads from multiple sources.
- Manage customer organizations.
- Track the complete sales pipeline.
- Schedule Discovery Calls.
- Record customer interactions.
- Generate proposals.
- Monitor sales opportunities.
- Convert approved opportunities into customer projects.
- Maintain a complete commercial history.

14.3 Scope

The module includes:

- Lead Management
- Opportunity Management
- Discovery Call Management
- Proposal Management

- Sales Pipeline
- Customer Timeline
- Activity Tracking
- Customer Notes
- Follow-up Management
- Project Handover
- Marketing campaign management

14.4 Actors

The following actors interact with this module:

- Admin
- Super Admin

Clients, Developers, and Technical Operators shall not have direct access to the CRM.

14.5 Lead Management

The platform shall support the creation of leads from:

- Public website contact forms.
- Discovery Call requests.
- AI Scan purchases.
- Partnership inquiries.
- Manual lead creation by Admins.
- Future external integrations.

Each lead shall have a unique record and shall remain associated with a single Organization once converted.

14.6 Lead Lifecycle

Every lead shall progress through a defined lifecycle.

The default lifecycle shall include:

- New
- Contacted
- Qualified
- Discovery Call Scheduled
- Discovery Call Completed
- Proposal In Progress
- Proposal Sent

- Negotiation
- Won
- Lost
- Archived
- Status changes shall be recorded in the audit log.

14.7 Discovery Call Management

The module shall enable Admins to:

- Schedule Discovery Calls.
- Reschedule calls.
- Cancel calls.
- Record meeting outcomes.
- Add meeting notes.
- Assign follow-up actions.
- Link Discovery Calls to lead records.

Each Discovery Call shall remain associated with its originating lead or customer organization.

14.8 Opportunity Management

Qualified leads may be converted into sales opportunities.

Each opportunity shall include:

- Customer organization.
- Estimated value.
- Expected close date.
- Current stage.
- Responsible Admin.
- Related Discovery Calls.
- Associated proposals.
- Opportunities shall remain linked to their originating lead history.

14.9 Proposal Management

Admins shall be able to:

- Create proposals.
- Update proposal status.
- Record customer responses.
- Track proposal versions.
- Mark proposals as accepted or rejected.

- Accepted proposals may initiate the customer onboarding workflow.

14.10 Customer Timeline

The CRM shall maintain a chronological history of all commercial activities, including:

- Lead creation.
- Contact attempts.
- Discovery Calls.
- Emails and messages.
- Proposal updates.
- Status changes.
- Notes.
- AI Scan purchases.
- Project handover.

The timeline shall provide a complete historical view for authorized users.

14.11 Activity Management

Admins shall be able to:

- Create follow-up tasks.
- Assign responsibilities.
- Schedule reminders.
- Record completed activities.
- View overdue actions.

Activities shall remain linked to the relevant lead, opportunity, or customer organization.

14.12 Project Handover

When an opportunity is marked as Won, the platform shall support the transition from Sales to Delivery.

The handover process shall:

- Create a customer project.
- Associate the project with the customer organization.
- Preserve all commercial history.
- Notify the appropriate Admins.
- Prepare the project for operational delivery.
- No commercial data shall be lost during this transition.

14.13 Business Rules

Rule	Business Rule
BR-006	Organization Isolation
BR-007	Data Ownership
BR-018	Project Ownership
BR-020	Audit Logging

14.14 Security Requirements

The CRM shall:

- Restrict access to authorized administrative users.
- Protect confidential commercial information.
- Record all customer record modifications.
- Maintain proposal history.
- Protect customer communication records.
- Verify permissions before displaying commercial data.

14.15 Acceptance Criteria

The module shall be considered complete when:

- Leads can be created and managed.
- Discovery Calls can be scheduled and tracked.
- Opportunities progress through the defined lifecycle.
- Proposals are managed successfully.
- Customer timelines display complete histories.
- Activities and follow-ups function correctly.
- Won opportunities create project handovers.
- All commercial activities are recorded in the audit log.

14.16 Automation Requirements

The CRM & Sales Management module shall maximize automation across the commercial lifecycle while ensuring that business-critical decisions remain under human control where required.

The platform shall:

Automatically capture leads from all supported entry points, including the Public Website,

Discovery Call requests, AI Scan purchases, partnership inquiries, and future integrated lead sources.

Automatically assign and route leads according to configurable business rules, ownership assignments, territories, or other administrative criteria.

Automatically create and maintain customer records, commercial activities, and audit entries throughout the sales lifecycle.

Automatically execute follow-up workflows, including reminders, task creation, email notifications, and status updates based on predefined business events.

Support automated email communication throughout the customer journey, including confirmations, reminders, onboarding communications, and other commercial interactions.

Support automated sales pipeline progression where predefined business conditions have been satisfied while allowing authorized users to override automated actions when necessary.

Support campaign management for commercial, marketing, and customer engagement activities as part of the integrated business ecosystem.

Automatically analyze leads using configurable business criteria together with AI-assisted evaluation to support lead qualification, prioritization, and commercial decision-making.

Ensure that AI-generated recommendations support, but do not replace, the decision-making responsibilities of authorized users.

Execute business workflows automatically whenever predefined conditions have been met, unless a Business Rule explicitly requires human review, approval, or authorization before the workflow may continue.

___

14.17 Out of Scope

The following functionality is excluded from

Version 1:

- External CRM synchronization.
- Commission management.
- Advanced sales forecasting.
- These capabilities may be introduced in future platform versions.

14.18 Developer Notes

This module defines the functional and business requirements for CRM & Sales Management.

The development team is responsible for implementing lead management, opportunity tracking, proposal workflows, customer timelines, and project handover functionality while ensuring compliance with all platform- wide business rules, security requirements, and audit obligations.

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MODULE 9

15. Module 9 — Project & Delivery Management

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

15.1 Purpose

The Project & Delivery Management module provides a centralized environment for planning, executing, monitoring, and successfully delivering customer projects.

This module ensures that every customer engagement follows a standardized delivery process with clearly defined responsibilities, milestones, approvals, and deliverables.

The objective is to provide operational transparency, predictable project execution, and high-quality service delivery while maintaining complete traceability throughout the project lifecycle.

15.2 Objectives

The Project & Delivery Management module shall enable authorized users to:

- Create customer projects.
- Plan project phases.
- Define milestones and deliverables.
- Assign resources.
- Manage tasks.
- Monitor project progress.
- Record project decisions.
- Manage project documentation.
- Track project risks and issues.
- Complete customer approvals.
- Successfully close projects.

15.3 Scope

The module includes:

- Project Creation
- Project Planning

- Project Lifecycle
- Task Management
- Milestone Management
- Deliverable Management
- Resource Assignment
- Risk & Issue Management
- Customer Approvals
- Project Documentation
- Project Completion
- Project History

15.4 Actors

The following actors interact with this module:

- Admin
- Super Admin
- Developer
- Client (limited access through the Client
- Portal)

Technical Operators may access project information only where explicitly authorized for technical support purposes.

15.5 Project Lifecycle

Every project shall progress through a standardized lifecycle.

The default lifecycle shall include:

- Created
- Planning
- Kick-off
- In Progress
- Internal Review
- Customer Review
- Approved
- Completed
- Closed
- Archived

Projects shall not skip mandatory approval stages unless explicitly authorized according to platform business rules.

15.6 Project Planning

Authorized users shall be able to:

- Define project objectives.
- Create project phases.
- Define milestones.
- Establish dependencies.
- Estimate timelines.
- Assign responsibilities.
- Define expected deliverables.
- Update planning where required.
- All planning changes shall be recorded in the audit log.

15.7 Task Management

The platform shall support structured task management.

Authorized users shall be able to:

- Create tasks.
- Assign tasks.
- Define priorities.
- Set due dates.
- Track progress.
- Add comments.
- Attach files.
- Mark tasks as completed.
- Reopen tasks where necessary.

Task ownership shall remain clearly identifiable throughout the project lifecycle.

15.8 Milestone Management

Projects shall support milestone-based delivery.

Each milestone shall include:

- Title.
- Description.
- Target completion date.
- Responsible owner.
- Associated deliverables.
- Current status.

- Approval status.

Milestones requiring customer approval shall not be marked as completed until approval has been recorded.

15.9 Deliverable Management

Each project may contain one or more deliverables.

Authorized users shall be able to:

- Upload deliverables.
- Replace deliverables.
- Maintain version history.
- Submit deliverables for review.
- Approve deliverables.
- Archive completed deliverables.
- Published deliverables shall become available to
- Clients through the Client Portal where applicable.

15.10 Resource Management

Authorized users shall be able to:

- Assign Developers.
- Change project assignments.

Change projects name.(for client privacy when assign project to a Developer)

Review resource availability.

Monitor workload.

Remove assignments where necessary.

Resource allocation shall support future project scaling.

15.11 Risk & Issue Management

The platform shall support structured management of project risks and issues.

Authorized users shall be able to:

- Register risks.
- Register issues.
- Assign ownership.
- Define mitigation actions.

- Track resolution progress.
- Close resolved items.
- Historical records shall remain available after project completion.

15.12 Customer Approval

Workflow

Customer approval shall be supported for designated project phases and deliverables.

The approval workflow shall:

- Notify the customer.
- Record approval decisions.
- Record rejection reasons where applicable.
- Trigger follow-up workflows.
- Record audit events.

Projects shall not advance beyond mandatory approval stages without the required customer approval.

15.13 Project Documentation

Each project shall maintain a centralized repository for project-related documentation, including:

- Requirements.
- Specifications.
- Deliverables.
- Technical documentation.
- Customer documentation.
- Approval records.
- Meeting notes.
- Supporting files.
- Documentation shall remain associated with the owning Organization.

15.14 Project Completion

A project may only be completed when:

- All mandatory milestones have been completed.
- Required approvals have been obtained.
- Outstanding critical issues have been resolved or formally accepted.
- Final deliverables have been published where applicable.
- Completion has been confirmed by an authorized Admin.

Upon completion, the platform shall archive the project history while maintaining future access according to authorization policies.

15.15 Business Rules

Rule	Business Rule
BR-006	Organization Isolation
BR-007	Data Ownership
BR-018	Project Ownership
BR-019	Approval Workflow
BR-020	Audit Logging
BR-022	Event-Driven Notifications

15.16 Automation Requirements

The platform shall automatically:

Create a project after a sales opportunity has been marked as Won, where configured.

Notify assigned team members of new project assignments.

Predictive project forecasting using AI.

Generate project timelines based on predefined templates where applicable.

Notify responsible users of upcoming milestones and overdue tasks.

Record project history and status changes.

Update dashboards and operational metrics following project events.

Trigger customer notifications when approvals are required.

Archive completed projects according to platform policies.

Manual intervention shall only be required where business rules explicitly require human approval or decision-making.

15.17 Security Requirements

The Project & Delivery Management module shall:

- Restrict project access according to assigned roles and permissions.
- Verify Organization ownership before granting access.

- Protect confidential project documentation.
- Record all business-critical project actions.
- Prevent unauthorized modification of approved deliverables.

15.18 Acceptance Criteria

The module shall be considered complete when:

- Projects can be created and managed successfully.
- Tasks, milestones, and deliverables function as specified.
- Customer approvals operate correctly.
- Project documentation is securely managed.
- Automated workflows execute correctly.
- Notifications are delivered successfully.
- Audit events are recorded for all business- critical actions.
- Completed projects are archived according to platform policies.

15.19 Out of Scope

The following functionality is excluded from

Version 1:

- Portfolio management across multiple business entities.
- Automated resource optimization.
- External project management platform synchronization.
- These capabilities may be introduced in future platform versions.

15.20 Developer Notes

This module defines the business requirements for project planning, execution, collaboration, approval, and delivery.

The development team is responsible for implementing the project management workflows, task handling, document management, approval processes, automation mechanisms, and reporting capabilities while ensuring compliance with all platform-wide business rules and security requirements.

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MODULE 10

16. Module 10 — Commercial Billing & Subscription Management

Version	1.1	Priority	Critical
Status	Release Candidate	Business Owner	IO SKY

16.1 Purpose

The Billing & Financial Management module provides centralized management of all commercial transactions, invoicing, payments, subscriptions, and financial records within the IO

SKY Platform.

The module ensures that all financial activities are securely processed, accurately recorded, fully auditable, and integrated with the platform's operational workflows.

16.2 Objectives

The Billing & Financial Management module shall enable authorized users to:

- Generate quotations.
- Convert approved quotations into invoices.
- Process payments.
- Manage subscriptions.
- Track payment status.
- Generate financial records.
- Process refunds where authorized.
- Maintain complete financial history.
- Support financial reporting.

16.3 Scope

The module includes:

- Quote Management
- Commercial Product Management
- Invoice Management

- Payment Management
- Subscription Management
- Payment Status Tracking
- Financial History
- Financial Reporting
- Customer Billing Overview
- Accounting software integration is outside the scope of Version 1.

16.4 Actors

The following actors interact with this module:

- Client
- Admin
- Super Admin

Technical Operators and Developers shall not access financial information unless explicitly authorized.

16.5 Quote Management

Authorized users shall be able to:

- Create quotations.
- Edit quotations.
- Send quotations.
- Track quotation status.
- Record customer responses.
- Convert accepted quotations into customer projects where applicable.
- Each quotation shall maintain version history.

16.6 Invoice Management

The platform shall support:

- Invoice generation.
- Invoice numbering.
- Invoice status tracking.
- Invoice downloads.
- Invoice history.
- Credit invoice generation where authorized.

Invoices shall remain permanently associated with the owning Organization.

16.7 Payment Management

The platform shall support secure payment processing.

The module shall:

- Verify payment completion.
- Record payment status.
- Record transaction references.
- Update related business workflows.
- Notify relevant users.

Business services shall not become active until payment confirmation has been received.

16.8 Subscription Management

The platform shall support recurring commercial services.

Authorized users shall be able to:

- View active subscriptions.
- Upgrade subscriptions.
- Downgrade subscriptions.
- Cancel subscriptions.
- View subscription history.
- Review renewal dates.
- Subscription changes shall be recorded in the audit log.

16.9 Commercial Products & Subscription Model

The IO SKY Platform shall support multiple commercial product types to accommodate different customer journeys and business models.

The platform shall distinguish between one-time products, implementation services, recurring subscriptions, and future commercial offerings.

Each commercial product shall define its own pricing model, billing method, activation requirements, and lifecycle.

The platform architecture shall support the addition of new commercial products without requiring fundamental changes to the billing system.

16.9.1 One-Time Products

The platform shall support products that are purchased as a single transaction.

Examples include:

- AI Scan Reports
- Custom Software Projects
- Professional Services
- Consulting Engagements
- Future one-time commercial offerings

One-time products shall not automatically create recurring billing unless explicitly configured.

16.9.2 Subscription Products

The platform shall support recurring subscription-based services.

Subscription products may include operational services, monitoring services, security services, managed services, AI-powered services, or future recurring offerings defined by IO SKY.

Each subscription shall define:

- Billing frequency
- Activation conditions
- Service period
- Renewal policy
- Cancellation policy
- Upgrade options
- Downgrade options

The exact commercial subscription catalogue shall be configurable by the platform administrators.

16.9.3 Commercial Lifecycle

Commercial products may progress through the following lifecycle where applicable:

- Product Selection
- Quotation
- Customer Acceptance
- Payment Confirmation
- Service Activation
- Service Delivery
- Subscription Management (where applicable)
- Service Completion or Cancellation
- The exact workflow shall depend on the selected commercial product.
- 16.9.4 Future Product Expansion

The commercial architecture shall support future product categories without requiring redesign of the billing infrastructure.

Future product types may include:

- Additional AI services
- Managed operational services
- Premium subscriptions
- Enterprise services
- Marketplace offerings
- Other commercial products introduced by IO
- SKY
- The commercial product model shall remain fully extensible.
- _____

16.10 Financial History

The platform shall maintain a complete financial history for each Organization, including:

- Quotations.
- Invoices.
- Payments.
- Subscription changes.
- Financial adjustments.
- Financial records shall not be deleted.

16.11 Financial Reporting

Authorized users shall be able to view:

- Revenue summaries.
- Outstanding invoices.
- Payment completion rates.
- Subscription metrics.
- Financial trends.
- Reports shall support operational and executive decision-making.

16.12 Business Rules

Rule	Business Rule
BR-007	Data Ownership
BR-020	Audit Logging
BR-022	Event-Driven Notifications
BR-023	Payment Confirmation

16.13 Automation Requirements

The platform shall automatically:

- Generate invoices when required business conditions are met.
- Record successful payments.
- Update invoice status following payment confirmation.
- Activate purchased services after successful payment verification.
- Generate customer payment confirmations.
- Notify authorized Admins of failed or overdue payments.
- Generate renewal reminders for recurring subscriptions.
- Update dashboards and financial metrics following financial events.
- Record all financial transactions in the audit log.

Automatically activate purchased one-time products after successful payment verification.

Automatically activate recurring subscriptions when all activation conditions have been satisfied.

Automatically suspend, expire, or terminate subscriptions according to configurable business rules where applicable.

Manual intervention shall only be required where business policies require approval, review, or financial authorization.

16.14 Security Requirements

The Billing & Financial Management module shall:

- Restrict access to authorized users only.
- Protect confidential financial information.
- Encrypt sensitive payment-related data where applicable.
- Record all financial modifications.
- Prevent unauthorized changes to completed financial records.
- Maintain complete traceability of financial transactions.

16.15 Acceptance Criteria

The module shall be considered complete when:

- Quotations can be created and managed.
- Invoices are generated correctly.
- Payments are processed successfully.
- Payment confirmation activates the appropriate business workflows.
- Subscription management functions correctly.

- Financial reports display accurate information.
- Audit records are created for all financial events.

16.16 Out of Scope

The following functionality is excluded from

Version 1:

- Full accounting and bookkeeping.
- Payroll management.
- Banking administration.
- Multi-company accounting.
- These capabilities may be supported through future integrations.

16.17 Developer Notes

This module defines the business requirements for financial management within the IO SKY

Platform.

The development team is responsible for implementing quotation workflows, invoicing, payment processing, subscription management, financial reporting, and financial security while ensuring compliance with applicable business rules, audit requirements, and payment provider standards.

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MODULE 11

17. Module 11 — Notifications & Communication

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

17.1 Purpose

The Notifications & Communication module provides centralized management of all platform-generated communications, ensuring that users receive timely, relevant, and secure information regarding business events, platform activities, and operational updates.

This module establishes a standardized communication framework across the entire IO

SKY Platform while maintaining consistency, traceability, and user-specific preferences.

17.2 Objectives

The Notifications & Communication module shall enable the platform to:

- Deliver automated notifications.
- Support in-app communications.
- Deliver email communications.
- Support AI-powered communications.
- Support AI Receptionists.
- Support AI Support Agents.
- Support AI Sales Agents.
- Support AI Voice Agents.
- Support human-to-user communications where applicable.
- Notify users of business events.
- Maintain communication history.
- Support notification preferences.
- Record communication delivery status.

Support omnichannel communication across present and future communication channels.

17.3 Scope

The module includes:

- In-App Notifications
- Email Communications
- AI Agent Communications
- AI Receptionist Communications
- AI Support Communications
- AI Sales Communications
- AI Voice Communications
- System Notifications
- Business Event Notifications
- Notification Preferences
- Communication Templates
- Communication History
- Delivery Status Tracking
- Omnichannel Communication Infrastructure

17.4 Actors

The following actors interact with this module:

- Client
- Developer
- Admin
- Super Admin
- Technical Operator

Every authenticated user may receive notifications according to their assigned role and permissions.

17.5 Notification Categories

The platform shall support the following notification categories:

- Account Notifications
- Examples include:
- Account activation.
- Password reset.
- MFA updates.
- Login alerts.
- Session security notifications.
- AI Scan Notifications
- Examples include:
- Purchase confirmation.
- Questionnaire available.

- Questionnaire submitted.
- AI processing started.
- Expert review in progress.
- Report published.
- Project Notifications
- Examples include:
- Project created.
- Task assigned.
- Milestone reached.
- Approval requested.
- Deliverable available.
- Project completed.
- Financial Notifications
- Examples include:
- Quote available.
- Invoice issued.
- Payment confirmed.
- Payment failed.
- Subscription renewal.
- Refund completed.
- Administrative Notifications
- Examples include:
- User invitation.
- Permission changes.
- Platform announcements.
- Security updates.
- Maintenance notifications.

17.6 Communication Channels

Version 1 shall support communication through:

- In-App Notifications
- Email
- WhatsApp Business for emergency(only for existing clients)
- IVR & own telephone system(SIP)
- AI Receptionists
- AI Support Agents
- AI Sales Agents

- AI Voice Agents
- AI personalized email follow ups

The platform architecture shall additionally support integration with communication providers including, but not limited to:

- Telephone Providers
- SMS Providers
- WhatsApp Business
- Microsoft Teams
- Push Notifications
- Future communication providers selected by
- IO SKY

Communication channels shall remain fully configurable by platform administrators.

17.7 Notification Preferences

Users shall be able to configure notification preferences where permitted.

Preferences may include:

- Email notifications.
- In-app notifications.
- Security notifications.
- Project updates.
- Financial updates.
- Marketing communications.
- Platform-critical security notifications may not be disabled.

17.8 Notification Templates

The platform shall support reusable templates for:

- Emails.
- In-app notifications.
- System messages.
- Templates shall support:
- Dynamic placeholders.
- Localization.
- Version control.
- Administrative updates.

17.9 Communication History

The platform shall maintain a complete history of notifications and communications, including:

- Delivery timestamp.
- Recipient.
- Notification type.
- Delivery channel.
- Delivery status.
- Related business event.
- Communication history shall remain searchable for authorized users.

17.10 Business Rules

Rule	Business Rule
BR-020	Audit Logging
BR-022	Event-Driven Notifications

Notifications shall only be generated in response to valid business events or authorized administrative actions.

17.11 Automation Requirements

The platform shall automatically:

- Generate notifications following predefined business events.
- Deliver notifications through the appropriate communication channel.
- Route communications to AI Agents where configured.
- Trigger AI Receptionists where business workflows require.
- Trigger AI Support Agents for support workflows.
- Trigger AI Sales Agents for commercial workflows.
- Trigger AI Voice Agents for voice interactions where configured.
- Record delivery status.
- Retry failed deliveries according to configurable policies.
- Update communication history.
- Execute follow-up communications according to business rules.

Manual communication creation shall remain available for authorized administrative users where required.

17.12 Security Requirements

The Notifications & Communication module shall:

- Protect confidential communications.
- Deliver notifications only to authorized recipients.
- Prevent unauthorized access to communication history.
- Record notification delivery events.
- Protect communication templates from unauthorized modification.

17.13 Acceptance Criteria

The module shall be considered complete when:

- Notifications are generated for all defined business events.
- Email and in-app notifications are delivered successfully.
- User notification preferences are respected.
- Communication history is maintained.
- Notification templates function correctly.
- Delivery failures are recorded.

Security notifications are always delivered according to platform policy.

17.14 Out of Scope

The following functionality is excluded from

Version 1 implementation but shall remain supported by the platform architecture for future expansion:

- Public community chat platforms.
- Public social media messaging integrations.
- Marketing broadcast campaigns.
- Customer communication providers not yet selected by IO SKY.

The communication architecture shall remain fully extensible without requiring architectural redesign.

17.15 Notification Matrix

The Notification Matrix defines all mandatory platform notifications, their triggering events, recipients, delivery channels, automation behavior, and related entities required for platform implementation.

Notif. ID	Business Event	Trigger	Recipient Role(s)	Channel(s)	Auto / Manual	Template ID	Related Entity	Audit Req.	Retry / Escalation Tier
NT-001	Account Created	On event	User	Email	Automatic	—	User / Account	Yes	Standard (Tier 1)
NT-002	Account Activated	On event	User	Email + In-App	Automatic	—	User / Account	Yes	Standard (Tier 1)
NT-003	Password Reset Requested	On event	User	Email	Automatic	—	User / Account	Yes	Critical (Tier 2)
NT-004	Password Changed	On event	User	Email	Automatic	—	User / Account	Yes	Standard (Tier 1)
NT-005	MFA Enabled	On event	User	Email + In-App	Automatic	—	User / Account	Yes	Standard (Tier 1)
NT-006	Login From New Device	On event	User	Email + In-App	Automatic	—	User / Account	Yes	Standard (Tier 1)
NT-007	AI Scan Purchased	On event	Client	Email	Automatic	—	AI Scan / Report	Yes	Standard (Tier 1)
NT-008	Questionnaire Ready	On event	Client	Email + In-App	Automatic	—	AI Scan / Report	Yes	Standard (Tier 1)
NT-009	Questionnaire Submitted	On event	Client	In-App	Automatic	—	AI Scan / Report	Yes	Standard (Tier 1)
NT-010	AI Analysis Started	On event	Client	In-App	Automatic	—	AI Scan / Report	Yes	Standard (Tier 1)
NT-011	Expert Review Started	On event	Client	In-App	Automatic	—	AI Scan / Report	Yes	Standard (Tier 1)
NT-012	Report Approved	On event	Client	Email + In-App	Automatic	—	AI Scan / Report	Yes	Standard (Tier 1)
NT-013	Report Published	On event	Client	Email + In-App	Automatic	—	AI Scan / Report	Yes	Critical (Tier 2)
NT-014	Quote Created	On event	Client	Email	Automatic	—	Quote / Invoice / Subscription	Yes	Standard (Tier 1)
NT-015	Quote Accepted	On event	Admin	In-App	Automatic	—	Quote / Invoice / Subscription	Yes	Standard (Tier 1)
NT-016	Invoice Issued	On event	Client	Email	Automatic	—	Quote / Invoice / Subscription	Yes	Critical (Tier 2)
NT-017	Payment Successful	On event	Client, Admin	Email + In-App	Automatic	—	Quote / Invoice / Subscription	Yes	Critical (Tier 2)
NT-018	Payment Failed	On event	Client, Admin	Email + In-App	Automatic	—	Quote / Invoice / Subscription	Yes	Critical (Tier 2)
NT-019	Subscription Activated	On event	Client	Email + In-App	Automatic	—	Quote / Invoice / Subscription	Yes	Standard (Tier 1)
NT-020	Subscription Renewed	On event	Client	Email	Automatic	—	Quote / Invoice / Subscription	Yes	Standard (Tier 1)
NT-021	Subscription Cancelled	On event	Client, Admin	Email + In-App	Automatic	—	Quote / Invoice / Subscription	Yes	Standard (Tier 1)
NT-022	Project Created	On event	Assigned Team	In-App	Automatic	—	Project / Task	Yes	Standard (Tier 1)
NT-023	Project Assigned	On event	Assigned User	Email + In-App	Automatic	—	Project / Task	Yes	Standard (Tier 1)
NT-024	Task Assigned	On event	Assigned User	In-App	Automatic	—	Project / Task	Yes	Standard (Tier 1)
NT-025	Task Due Soon	On event	Assigned User	In-App	Automatic	—	Project / Task	Yes	Standard (Tier 1)
NT-026	Task Overdue	On event	User, Admin	In-App	Automatic	—	Project / Task	Yes	Standard (Tier 1)

Notif. ID	Business Event	Trigger	Recipient Role(s)	Channel(s)	Auto / Manual	Template ID	Related Entity	Audit Req.	Retry / Escalation Tier
NT-027	Customer Approval Required	On event	Client	Email + In-App	Automatic	—	Project / Task	Yes	Standard (Tier 1)
NT-028	Project Completed	On event	Client, Admin	Email + In-App	Automatic	—	Project / Task	Yes	Standard (Tier 1)
NT-029	AI Receptionist Assigned	On event	Client	AI Receptionist	Automatic	—	AI Agent Session	Yes	Standard (Tier 1)
NT-030	AI Support Session Started	On event	Client	AI Support Agent	Automatic	—	AI Agent Session	Yes	Standard (Tier 1)
NT-031	AI Sales Conversation Started	On event	Client	AI Sales Agent	Automatic	—	AI Agent Session	Yes	Standard (Tier 1)
NT-032	AI Voice Call Initiated	On event	Client	AI Voice Agent	Automatic	—	AI Agent Session	Yes	Standard (Tier 1)
NT-033	AI Escalation Required	On event	Admin	In-App	Automatic	—	AI Agent Session	Yes	Standard (Tier 1)
NT-034	AI Conversation Completed	On event	Admin	In-App	Automatic	—	AI Agent Session	Yes	Standard (Tier 1)
NT-035	User Invited	On event	User	Email	Automatic	—	Platform / User	Yes	Critical (Tier 2)
NT-036	Permission Changed	On event	Affected User	Email + In-App	Automatic	—	Platform / User	Yes	Standard (Tier 1)
NT-037	Platform Maintenance	On event	All Users	Email + In-App	Automatic	—	Platform / User	Yes	Standard (Tier 1)
NT-038	Security Incident	On event	Super Admin	Email + In-App	Automatic	—	Platform / User	Yes	Critical (Tier 2)
NT-039	Platform Announcement	On event	Target Audience	In-App	Manual	—	Platform / User	Yes	Standard (Tier 1)
NT-040	Discovery Call Confirmation	On event	Client	Email + In-App	Automatic	—	Lead / Discovery Call	Yes	Critical (Tier 2)
NT-041	Contract Approval	On event	Client, Admin	Email + In-App	Automatic	—	Document / Contract	Yes	Critical (Tier 2)

Trigger corresponds to the occurrence of the Business Event. Audit Required applies to every notification in accordance with BR-020. Retry/Escalation Tier is defined in full below (approved Product Owner decision, OPD-002).

Template ID: pending Product Owner approval, see Appendix D — Open Product Owner Decisions *(OPD-001).*

Retry and escalation behavior for failed notification delivery shall follow the tier assigned to each notification in the table above.

Every notification, regardless of tier, shall be retried on delivery failure according to the following schedule:

- Attempt 1: immediate.
- Attempt 2: after 5 minutes.
- Attempt 3: after 30 minutes.
- Attempt 4 (final): after 2 hours.

Standard (Tier 1) notifications — if all retry attempts fail:

- The notification shall be marked Failed.

- A record shall be created in the Audit Log.
- No escalation shall occur.

Critical (Tier 2) notifications — if all retry attempts fail:

- The notification shall be marked Failed.
- A record shall be created in the Audit Log.
- An Admin shall be notified.
- An Admin Dashboard Alert shall be displayed.

The following notifications are classified as Critical (Tier 2):

- Payment Confirmation (Payment Successful)
- Payment Failure (Payment Failed)
- AI Scan Completed (Report Published)
- Discovery Call Confirmation
- User Invitation (User Invited)
- Password Reset (Password Reset Requested)
- Security Alert (Security Incident)
- Contract Approval
- Invoice Delivery (Invoice Issued)

All other notifications in the Notification Matrix are classified as Standard (Tier 1). No additional notification tiers shall be introduced.

Every notification or communication generated by the platform shall create an immutable audit record containing:

- Notification Identifier
- Business Event
- Trigger Timestamp
- Recipient(s)
- Delivery Channel
- Delivery Status
- Related Entity
- Initiating User or Automated Process

17.16 Developer Notes

This module defines the functional communication requirements for the IO SKY

Platform.

The development team is responsible for implementing notification delivery, template management, communication history, user preferences, and delivery tracking while ensuring compliance with all platform-wide business rules, security requirements, and audit obligations.

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MODULE 12

18. Module 12 — File & Document Management

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

18.1 Purpose

The File & Document Management module provides centralized, secure, and structured management of all files and documents within the IO SKY Platform.

The module ensures that documents remain organized, version-controlled, securely stored, and accessible only to authorized users while maintaining complete traceability throughout the document lifecycle.

18.2 Objectives

The File & Document Management module shall enable the platform to:

- Store documents securely.
- Upload and download files.
- Organize files and folders.
- Maintain document version history.
- Control document permissions.
- Associate documents with platform entities.
- Support document approvals.
- Archive documents.
- Maintain document audit history.

18.3 Scope

The module includes:

- File Upload
- File Download
- Document Storage
- Folder Management
- Version Management

- Document Permissions
- Document Approval
- Document Sharing
- Document Search
- Document Archiving
- Document History
- File Metadata Management

18.4 Actors

The following actors interact with this module:

- Client
- Developer
- Admin
- Super Admin
- Technical Operator (where explicitly authorized)

Access shall always be determined by role, permissions, and Organization ownership.

18.5 Supported Documents

The platform shall support document management for, including but not limited to:

- AI Scan Reports
- Project Documentation
- Technical Documentation
- Requirements Specifications
- Quotations
- Invoices
- Contracts
- Proposals
- Meeting Notes
- Deliverables
- User Manuals
- Images
- Videos
- Presentations
- Other business-related documents approved by IO SKY

18.6 Document Organization

The platform shall support logical organization of documents through configurable structures.

Documents may be associated with:

- Organization
- Client
- Project
- AI Scan
- Opportunity
- Invoice
- Subscription
- User
- Developer
- Internal Operations
- Other platform entities

Each document shall remain linked to its originating business entity.

18.7 Version Management

The platform shall maintain document version history.

Authorized users shall be able to:

- Upload new versions.
- View version history.
- Restore previous versions where permitted.
- Compare document revisions where applicable.

Previous versions shall remain available according to document retention policies.

18.8 Document Permissions

Access to documents shall be controlled through the platform authorization model.

Permissions shall support:

- View
- Download
- Upload
- Replace
- Approve
- Archive

- Delete (where permitted)

Permission inheritance shall follow Organization ownership and platform business rules.

18.9 Document Approval Workflow

The platform shall support document approval workflows.

Authorized users shall be able to:

- Submit documents for review.
- Approve documents.
- Reject documents.
- Request revisions.
- Record approval decisions.
- Maintain approval history.

Approved documents shall be protected from unauthorized modification.

18.10 Search & Retrieval

Authorized users shall be able to search documents using:

- Document Name
- Project
- Client
- Organization
- Category
- Tags
- Creation Date
- Modification Date
- Status
- Document Type
- Search results shall respect access permissions.

18.11 Automation Requirements

The platform shall automatically:

- Store documents generated by business workflows.
- Associate documents with related business entities.
- Maintain version history.
- Generate document metadata.
- Archive documents according to configurable retention policies.

- Notify users when documents require review or approval.
- Record all document activities within the audit log.
- Manual document management shall remain available where authorized.

18.12 Security Requirements

The File & Document Management module shall:

- Protect confidential documents.
- Restrict access according to roles and permissions.
- Verify Organization ownership before granting access.
- Encrypt stored documents where applicable.
- Record all document access and modifications.
- Prevent unauthorized modification of approved documents.

18.13 Business Rules

Rule	Business Rule
BR-006	Organization Isolation
BR-007	Data Ownership
BR-020	Audit Logging
BR-022	Event-Driven Notifications

18.14 Acceptance Criteria

The module shall be considered complete when:

- Documents can be uploaded and downloaded securely.
- Version history operates correctly.
- Permissions are enforced correctly.
- Approval workflows function as specified.
- Search returns authorized results only.
- Automated document workflows execute successfully.

Audit records are created for all business- critical document actions.

18.15 Out of Scope

The following functionality is excluded from

Version 1 implementation:

- Real-time collaborative document editing.

- External cloud storage synchronization.
- Optical Character Recognition (OCR).
- AI-powered document content analysis.
- Digital signature providers not selected by IO
- SKY.

The document architecture shall remain extensible to support future integrations without requiring fundamental redesign.

18.16 Document Lifecycle

Every document managed by the IO SKY

Platform shall follow a standardized lifecycle to ensure consistency, traceability, and governance throughout its existence.

The default document lifecycle shall include:

- Draft
- Under Review
- Approved
- Published
- Archived
- Deleted (where explicitly permitted by business rules)

Documents shall only transition between lifecycle stages through authorized actions or automated workflows defined by the platform.

Mandatory approval stages shall not be bypassed unless explicitly authorized according to applicable business rules.

18.17 Document Classification

The platform shall support standardized document classification to ensure appropriate handling of sensitive information.

The default classification levels shall include:

- Public
- Internal
- Confidential
- Restricted

Document classification shall determine the applicable security controls, visibility, sharing permissions, and access restrictions.

Only authorized users may modify the classification of a document.

18.18 File Validation

The platform shall validate all uploaded files before they are accepted into the system.

Validation shall include, where applicable:

- Supported file type verification.
- Maximum file size validation.
- File integrity verification.
- Duplicate file detection.
- File name validation.
- Malware and virus scanning.
- Upload completion verification.
- Files failing validation shall not be stored within the platform.

Validation rules shall remain configurable by authorized administrators.

18.19 Storage & Retention Policies

The platform shall manage documents according to configurable storage and retention policies.

The document management system shall support:

- Active storage.
- Archived storage.
- Configurable retention periods.
- Secure long-term storage.
- Controlled permanent deletion where permitted.

Documents shall remain available for the duration of their applicable retention period unless superseded by legal, regulatory, or contractual obligations.

18.20 Document Relationships

The platform shall support relationships between documents and multiple business entities.

A single document may be associated with one or more of the following:

- Organization
- Client
- Project
- AI Scan
- Opportunity

- Invoice
- Subscription
- User
- Developer
- Task
- Milestone
- Deliverable
- Support Case
- Internal Operation

The document relationship model shall remain extensible to support future business entities introduced within the platform.

18.21 Document Matrix

The Document Matrix defines the minimum governance requirements for all document types managed within the IO SKY Platform.

Each document type shall define:

- Document Type
- Owning Business Entity
- Authorized Roles
- Version Control Requirement
- Approval Requirement
- Retention Policy
- Archive Policy
- Classification Level

Example document categories include, but are not limited to:

Document Type	Owning Entity	Authorized Roles	Version Control Required	Approval Required	Default Classification	Retention Rule	Archive Rule
AI Scan Report	—	—	Yes	Yes	—	—	—
Project Specification	—	—	Yes	Yes	—	—	—
Technical Documentation	—	—	Yes	Yes	—	—	—
Project Deliverable	—	—	Yes	Yes	—	—	—
Proposal	—	—	Yes	Yes	—	—	—
Quotation	—	—	Yes	Yes	—	—	—
Invoice	—	—	No	No	—	—	—
Contract	—	—	Yes	Yes	—	—	—
Meeting Notes	—	—	Yes	No	—	—	—
User Manual	—	—	Yes	Yes	—	—	—
Internal Documentation	—	—	Yes	Optional	—	—	—

Version Control and Approval requirements are defined by this specification.

Owning Entity, Authorized Roles, Default Classification, Retention Rule and Archive Rule: pending Product Owner approval, see Appendix D — Open Product Owner Decisions *(OPD-003).*

The complete Document Matrix shall be configurable by authorized administrators while maintaining compliance with platform-wide governance, security, and audit requirements.

18.22 Business Continuity & Disaster Recovery

The platform shall ensure the integrity, availability, and recoverability of all managed documents.

The document management architecture shall support:

- Secure backup procedures.
- Disaster recovery processes.
- Document integrity verification.

Recovery of stored documents following system failures where technically feasible.

Protection against unauthorized data loss or corruption.

Document availability shall support the platform's operational continuity objectives.

———

18.23 Developer Notes

This module defines the business requirements for document storage, organization, version management, approval workflows, security, and lifecycle management.

The development team is responsible for implementing secure document management while ensuring compliance with platform-wide authorization, audit, workflow, and security requirements.

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MODULE 13

19. Module 13 — AI Intelligence Layer

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

19.1 Purpose

The AI Intelligence Layer provides the centralized artificial intelligence infrastructure for the IO SKY Platform.

This module enables AI-powered analysis, decision support, workflow automation, intelligent assistance, and autonomous execution while ensuring that all AI activities remain secure, transparent, auditable, and governed according to platform business rules.

The AI Intelligence Layer shall function as a shared platform service available to all authorized modules within the IO SKY Platform.

19.2 Objectives

The AI Intelligence Layer shall enable the platform to:

- Execute AI-powered workflows.
- Support AI Agents.
- Support AI Decision Support.
- Execute AI-assisted business processes.
- Generate AI recommendations.
- Analyze structured and unstructured data.
- Support human-in-the-loop approval workflows.
- Maintain AI audit history.
- Support configurable AI providers.
- Continuously improve operational intelligence.

19.3 Scope

The module includes:

- AI Orchestration
- AI Agents

- AI Workflows
- AI Decision Support
- AI Prompt Management
- AI Model Management
- AI Context Management
- AI Memory
- AI Recommendations
- AI Approval Workflows
- AI Audit Logging
- AI Performance Monitoring

19.4 Actors

The following actors interact with this module:

- Client
- Developer
- Admin
- Super Admin
- Technical Operator
- AI Agents

AI Agents shall operate only within the permissions and business rules assigned by the platform.

19.5 AI Agent Framework

The platform shall support multiple specialized

AI Agents operating within a centralized orchestration framework.

Examples include, but are not limited to:

- AI Support Agent
- AI Sales Agent
- AI Receptionist
- AI Voice Agent
- AI Operations Agent
- AI Project Assistant
- AI Document Assistant
- AI Reporting Agent
- AI Compliance Assistant
- AI Executive Assistant

The platform architecture shall support the addition of future AI Agents without requiring architectural redesign.

19.6 AI Orchestration

The AI Intelligence Layer shall coordinate AI

Agents, business workflows, platform services, and external AI providers.

The orchestration engine shall:

- Route AI requests.
- Coordinate multiple AI Agents.
- Execute AI workflows.
- Manage AI context.
- Maintain execution history.
- Trigger business events.
- Support parallel AI execution where applicable.

19.7 AI Decision Support

The platform shall provide AI-generated recommendations to authorized users.

Examples include:

- Business recommendations.
- Operational recommendations.
- Workflow recommendations.
- Security recommendations.
- Project recommendations.
- Customer recommendations.

AI recommendations shall support human decision-making and shall not automatically override mandatory business approvals.

19.8 AI Memory & Context

Management

The platform shall maintain contextual information required for AI-assisted operations.

The AI Intelligence Layer shall support:

- Conversation context.
- Workflow context.
- Project context.

- Client context.
- Organization context.
- Historical interactions.
- Business rules.
- Permission-aware context retrieval.

AI Agents shall only access information that the requesting user is authorized to access.

19.9 AI Prompt Management

The platform shall support centralized management of AI prompts.

Authorized administrators shall be able to:

- Create prompts.
- Update prompts.
- Version prompts.
- Test prompts.
- Assign prompts to AI Agents.
- Archive obsolete prompts.
- Prompt changes shall be recorded within the audit log.

19.10 AI Workflow Automation

The AI Intelligence Layer shall automatically execute AI-powered workflows where configured.

Examples include:

- AI Scan analysis.
- Document summarization.
- Customer assistance.
- Lead qualification.
- Project recommendations.
- Risk analysis.
- Operational monitoring.
- Business intelligence generation.

Human approval shall remain mandatory where required by applicable business rules.

19.11 AI Security Requirements

The AI Intelligence Layer shall:

- Respect platform authorization.
- Respect Organization isolation.

- Prevent unauthorized data exposure.
- Record AI execution history.
- Protect confidential prompts.
- Protect AI-generated outputs.
- Maintain secure communication with external AI providers.

19.12 AI Governance

The platform shall ensure responsible AI usage.

Governance shall include:

- Human oversight.
- Explainability where applicable.
- Auditability.
- Permission enforcement.
- Configurable AI policies.
- Responsible AI decision support.

Business-critical decisions requiring human authorization shall never be executed solely by

AI unless explicitly configured by authorized administrators.

19.13 Business Rules

Rule	Business Rule
BR-006	Organization Isolation
BR-007	Data Ownership
BR-020	Audit Logging
BR-022	Event-Driven Notifications

19.14 Acceptance Criteria

The module shall be considered complete when:

- AI Agents operate according to assigned permissions.
- AI workflows execute successfully.
- AI recommendations are generated correctly.
- AI prompts are centrally managed.
- AI activities are audited.
- Human approval workflows function correctly.
- AI services integrate successfully with platform modules.

19.15 Out of Scope

The following functionality is excluded from

Version 1 implementation:

- Self-modifying AI systems.
- Autonomous AI policy creation.
- AI model training within the platform.
- AI systems operating outside platform governance.

Future AI capabilities shall remain compatible with the platform architecture.

19.16 Developer Notes

This module defines the business requirements for the centralized AI Intelligence Layer.

The development team is responsible for implementing AI orchestration, AI Agent execution, prompt management, context handling, workflow automation, governance, monitoring, and secure AI integration while ensuring compliance with all platform-wide business rules and security requirements.

19.17 AI Agent Registry

The platform shall maintain a centralized registry of all AI Agents operating within the IO

SKY Platform.

Each AI Agent shall include, at a minimum:

- Agent Identifier
- Agent Name
- Description
- Assigned Business Function
- Supported Modules
- Authorized Capabilities
- Available Tools
- Accessible Knowledge Sources
- Assigned Permissions
- Escalation Rules
- Operational Status
- Version Information
- Responsible Admin
- The AI Agent Registry shall support future AI
- Agents without requiring architectural redesign.

19.18 AI Capability & Permission Matrix

Each AI Agent shall operate according to a predefined capability and permission model.

The platform shall define for every AI Agent:

- Permitted Actions
- Restricted Actions
- Accessible Business Data
- Authorized Platform Modules
- Human Approval Requirements
- Maximum Operational Authority
- Organization Scope
- Security Restrictions

AI Agents shall never execute actions outside their assigned permissions.

Permission enforcement shall be verified before every AI action.

19.19 AI Tool Execution

The platform shall support secure execution of business actions initiated by AI Agents.

Supported AI tool execution may include:

- Creating CRM records
- Updating CRM records
- Scheduling meetings
- Creating tasks
- Updating projects
- Sending notifications
- Generating reports
- Creating documentation
- Managing workflows
- Executing AI Scan processes

Business-critical actions involving financial, legal, contractual, or security-sensitive operations shall require human approval where defined by platform business rules.

19.20 AI Knowledge Management

The AI Intelligence Layer shall provide secure access to organizational knowledge required for

AI-assisted operations.

Knowledge sources may include:

- AI Scan Reports
- Client Documentation
- Project Documentation
- Technical Documentation
- Internal Procedures
- Knowledge Base Articles
- Product Documentation
- Business Rules
- Operational Data
- Platform Configuration

AI Agents shall only retrieve information that the requesting user or AI Agent is authorized to access.

Knowledge retrieval shall comply with platform- wide authorization, organization isolation, and security requirements.

19.21 AI Conversation & Interaction Management

The platform shall maintain contextual management of AI interactions.

The AI Intelligence Layer shall support:

- Conversation History
- Context Preservation
- Multi-session Continuity
- Conversation Summaries
- User Identification
- Organization Context
- Interaction History
- Action History
- Conversation Recovery

Where applicable, AI conversations shall remain available across supported communication channels.

19.22 Human Escalation & Approval

The platform shall support seamless escalation from AI Agents to authorized human users.

Escalation shall occur where required by business rules or predefined operational policies.

Examples include:

- Legal matters
- Financial commitments
- Security incidents

- Customer complaints
- Contract modifications
- Low AI confidence
- Unauthorized requests
- Business-critical approvals
- Escalation history shall be recorded within the audit log.

19.23 AI Safety & Governance

The platform shall enforce standardized AI safety policies.

The AI Intelligence Layer shall provide controls for:

- Prompt Injection Protection
- Unauthorized Data Access Prevention
- Sensitive Information Protection
- AI Output Validation
- Tool Execution Validation
- Permission Enforcement
- Responsible AI Operation
- Human Oversight
- Auditability
- AI Policy Enforcement

AI Agents shall never bypass platform security, authorization, or business rules.

19.24 AI Monitoring & Performance

The platform shall continuously monitor AI performance.

Authorized administrators shall be able to review:

- AI Agent Activity
- Execution History
- Success Rate
- Failure Rate
- Escalation Frequency
- Average Response Time
- AI Provider Usage
- Operational Costs
- Token Consumption
- Human Intervention Frequency
- Workflow Completion Rate

- AI Availability

Monitoring data shall support operational optimization, performance analysis, and responsible AI governance.

19.25 AI Provider Management

The AI Intelligence Layer shall support integration with multiple AI providers and models.

Authorized administrators shall be able to configure:

- Available AI Providers
- Available AI Models
- Default Model Selection
- Provider Priority
- Model Selection Per Workflow
- Cost Controls
- Usage Limits
- Fallback Providers
- Provider Availability Monitoring
- Model Version Management

The platform shall support switching between AI providers without requiring modifications to business workflows or platform architecture.

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MODULE 14

20. Module 14 — Audit, Compliance & Security Monitoring

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

20.1 Purpose

The Audit, Compliance & Security Monitoring module provides centralized governance, traceability, compliance monitoring, and security oversight across the entire IO SKY Platform.

The module ensures that all business-critical activities are recorded, monitored, and protected while supporting regulatory compliance, operational transparency, incident investigation, and platform security.

This module shall function as a shared platform service used by every module within the IO SKY

Platform.

20.2 Objectives

The Audit, Compliance & Security Monitoring module shall enable the platform to:

- Record business-critical activities.
- Monitor security events.
- Maintain immutable audit logs.
- Support compliance requirements.
- Detect operational anomalies.
- Monitor platform integrity.
- Record security incidents.
- Support forensic investigations.
- Maintain historical event records.
- Provide security reporting.

20.3 Scope

The module includes:

- Audit Logging
- Security Monitoring
- Compliance Monitoring
- Security Incident Management
- Event History
- Access Logging
- Authentication Logging
- Authorization Monitoring
- Platform Health Monitoring
- Audit Reporting
- Security Reporting
- Compliance Reporting

20.4 Actors

The following actors interact with this module:

- Super Admin
- Admin (where authorized)
- Technical Operator (where explicitly authorized)

Clients, Developers, and AI Agents shall not have direct access to platform-wide audit information unless explicitly authorized.

20.5 Audit Logging

The platform shall record all business-critical events.

Examples include:

- User authentication.
- Permission changes.
- AI Agent execution.
- Project updates.
- Document approvals.
- Payment confirmations.
- Subscription changes.
- Security events.
- Administrative actions.
- Platform configuration changes.
- Audit records shall be immutable.

20.6 Security Monitoring

The platform shall continuously monitor security- related events.

Examples include:

- Failed login attempts.
- Suspicious authentication.
- Unauthorized access attempts.
- Permission violations.
- AI policy violations.
- Data access anomalies.
- API misuse.
- Security alerts.

Security events shall be prioritized according to configurable severity levels.

20.7 Compliance Monitoring

The platform shall support compliance monitoring through configurable policies.

Compliance monitoring may include:

- Data access compliance.
- Audit completeness.
- Authorization compliance.
- Document retention compliance.
- AI governance compliance.
- Internal operational policies.
- Future regulatory requirements.

Compliance policies shall remain configurable by authorized administrators.

20.8 Security Incident

Management

Authorized users shall be able to:

- Register security incidents.
- Classify incident severity.
- Assign ownership.
- Track investigation progress.
- Record corrective actions.
- Close incidents.

- Maintain complete incident history.

All security incidents shall remain traceable throughout their lifecycle.

20.9 Event History

The platform shall maintain historical records of:

- Authentication events.
- Authorization events.
- AI activities.
- Business workflow execution.
- Administrative activities.
- Configuration changes.
- Security incidents.
- Compliance events.

Historical records shall remain searchable according to authorization policies.

20.10 Monitoring & Alerting

The platform shall automatically monitor business-critical activities.

Monitoring shall support:

- Security alerts.
- Operational alerts.
- AI monitoring alerts.
- Platform health alerts.
- Workflow failures.
- Integration failures.
- Compliance alerts.

Authorized users shall receive notifications according to the Notification Matrix.

20.11 Automation Requirements

The platform shall automatically:

- Record audit events.
- Monitor security events.
- Detect predefined anomalies.
- Generate security alerts.
- Generate compliance alerts.
- Record incident history.

- Notify authorized personnel.
- Maintain immutable audit records.

Manual intervention shall only be required where investigation, approval, or corrective action is necessary.

20.12 Security Requirements

The Audit, Compliance & Security Monitoring module shall:

- Protect audit integrity.
- Prevent unauthorized modification of audit records.
- Protect security information.
- Restrict access according to authorization.
- Record all administrative actions.
- Support forensic investigations.
- Maintain secure storage of audit records.

20.13 Business Rules

Rule	Business Rule
BR-006	Organization Isolation
BR-007	Data Ownership
BR-020	Audit Logging
BR-021	Audit Visibility
BR-022	Event-Driven Notifications

20.14 Acceptance Criteria

The module shall be considered complete when:

- Audit records are generated for all business- critical activities.
- Security monitoring operates continuously.
- Compliance monitoring functions correctly.
- Security incidents are managed successfully.
- Alerts are generated according to platform rules.
- Audit history remains immutable.
- Security reports are available to authorized users.

20.15 Out of Scope

The following functionality is excluded from

Version 1 implementation:

- External Security Information and Event
- Management (SIEM) integrations.
- Automated penetration testing.
- External compliance certification platforms.
- Third-party governance platforms not selected by IO SKY.

The architecture shall remain extensible to support future security and compliance integrations.

20.16 Developer Notes

This module defines the business requirements for audit logging, security monitoring, compliance oversight, incident management, and operational governance.

The development team is responsible for implementing secure audit storage, monitoring capabilities, compliance controls, alerting mechanisms, reporting, and authorization while ensuring alignment with all platform-wide business rules and security requirements.

20.17 Audit Event Matrix

The platform shall maintain a standardized Audit

Event Matrix defining all business-critical events that require mandatory audit logging.

Each audit event shall specify:

- Event Identifier
- Business Event
- Module
- Trigger
- Initiating User or System Process
- Related Business Entity
- Timestamp
- Audit Severity
- Retention Policy

Examples of mandatory audit events include, but are not limited to:

- User Authentication
- Failed Authentication
- Password Reset

- Permission Changes
- User Creation
- User Deletion
- AI Agent Execution
- AI Agent Escalation
- AI Prompt Modification
- AI Workflow Execution
- AI Configuration Changes
- Project Creation
- Project Completion
- Task Assignment
- Document Upload
- Document Approval
- Invoice Generation
- Payment Confirmation
- Subscription Activation
- Platform Configuration Changes
- Security Incidents

Audit requirements shall remain extensible for future platform modules.

20.18 Incident Severity Matrix

The platform shall classify all operational and security incidents according to standardized severity levels.

The default severity levels shall include:

- Critical
- Examples include:
- Data breach
- Unauthorized administrative access
- Platform-wide outage
- Security compromise
- Financial integrity failure
- High
- Examples include:
- Repeated unauthorized access attempts
- AI policy violations
- Service disruption
- Failed business-critical workflows

- Medium
- Examples include:
- Failed integrations
- Configuration inconsistencies
- Operational warnings
- Low
- Examples include:
- Minor validation errors
- Informational operational events
- Non-critical workflow failures
- Each severity level shall define:
- Notification Requirements
- Escalation Rules
- Response Priority
- Investigation Requirements
- Closure Requirements

Severity definitions shall remain configurable by authorized administrators.

20.19 Compliance Control Matrix

The platform shall maintain a centralized

Compliance Control Matrix defining mandatory governance controls applicable across all modules.

Each control shall define:

- Control Identifier
- Business Rule Reference
- Applicable Module(s)
- Validation Method
- Monitoring Method
- Compliance Status
- Responsible Role
- Compliance controls shall include, but are not limited to:
- Organization Isolation
- Data Ownership
- Authorization Enforcement
- Audit Logging
- Workflow Automation
- AI Governance

- Document Governance
- Financial Governance
- Security Monitoring
- Notification Governance

Compliance validation shall be performed continuously where technically feasible.

20.20 Security Policy Management

The platform shall support centralized management of security policies.

Authorized administrators shall be able to configure:

- Authentication Policies
- Password Policies
- Multi-Factor Authentication Policies
- Session Policies
- Access Control Policies
- AI Security Policies
- Data Protection Policies
- Incident Response Policies
- Policy modifications shall be recorded within the audit log.

20.21 Threat Detection & Response

The platform shall continuously monitor for suspicious or unauthorized activities.

Threat detection shall include, where applicable:

- Suspicious Login Activity
- Privilege Escalation Attempts
- Unauthorized Data Access
- AI Policy Violations
- Unusual User Behaviour
- API Misuse
- Configuration Manipulation
- Excessive Failed Requests

Detected threats shall generate alerts according to the Incident Severity Matrix.

20.22 Data Retention & Evidence Management

The platform shall securely retain audit records, incident records, compliance evidence, and security events according to configurable retention policies.

Retention management shall support:

- Audit Record Retention
- Incident Record Retention
- Compliance Evidence
- Investigation Records
- Secure Archiving
- Controlled Deletion where permitted

Business-critical records shall remain immutable throughout their retention period.

20.23 Security Dashboard

Authorized users shall have access to centralized security and compliance dashboards.

The dashboards shall provide visibility into:

- Active Security Alerts
- Open Incidents
- Incident Severity Distribution
- Compliance Status
- Audit Activity
- Failed Authentication Attempts
- AI Security Events
- Platform Health
- Operational Risks
- Recent Administrative Activities

Dashboard visibility shall be determined by user role and assigned permissions.

20.24 Business Continuity & Disaster Recovery

The platform shall support business continuity through standardized recovery procedures.

The architecture shall support:

- Backup Management
- Disaster Recovery Procedures
- System Recovery Validation
- Platform Availability Monitoring
- Critical Service Recovery
- Data Integrity Verification
- Operational Continuity Planning

Business continuity processes shall support the recovery objectives established by IO SKY.

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MODULE 15

21. Module 15 — Analytics & Business Intelligence

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

21.1 Purpose

The Analytics & Business Intelligence module provides centralized reporting, operational analytics, executive dashboards, and business intelligence across the IO SKY Platform.

The module enables authorized users to transform operational data into actionable insights that support informed decision-making, performance optimization, and strategic growth.

This module shall function as a shared platform service available across all platform modules.

21.2 Objectives

The Analytics & Business Intelligence module shall enable the platform to:

- Present operational dashboards.
- Display executive KPIs.
- Analyze business performance.
- Monitor AI performance.
- Track financial metrics.
- Analyze project performance.
- Measure operational efficiency.
- Generate configurable reports.
- Support data-driven decision-making.
- Provide historical trend analysis.

21.3 Scope

The module includes:

- Executive Dashboards
- Operational Dashboards
- Business KPIs
- Financial Analytics

- Project Analytics
- AI Analytics
- Client Analytics
- Sales Analytics
- Performance Reports
- Historical Trend Analysis
- Custom Dashboards
- Data Export

21.4 Actors

The following actors interact with this module:

- Client (limited dashboards)
- Developer (where authorized)
- Admin
- Super Admin
- Technical Operator (where authorized)

Dashboard visibility shall always be determined by user role, permissions, and Organization ownership.

21.5 Executive Dashboards

The platform shall provide executive dashboards presenting high-level business information.

Examples include:

- Revenue Overview
- Active Clients
- Active Projects
- AI Scan Statistics
- AI Agent Activity
- Subscription Overview
- Operational Health
- Security Status
- Outstanding Tasks
- Platform Growth
- Executive dashboards shall support configurable widgets.

21.6 Operational Dashboards

Operational dashboards shall support day-to-day management.

Examples include:

- Active Projects
- Pending Approvals
- Support Activity
- AI Workflows
- Document Activity
- User Activity
- Notifications
- System Events
- Workflow Performance

Operational dashboards shall update automatically where technically feasible.

21.7 Key Performance Indicators

(KPIs)

The platform shall support configurable KPIs.

Examples include:

- Revenue
- Monthly Recurring Revenue (MRR)
- Annual Recurring Revenue (ARR)
- Active Clients
- AI Scan Conversion Rate
- Project Completion Rate
- Customer Satisfaction
- Support Response Time
- AI Success Rate
- Workflow Automation Rate
- KPIs shall remain configurable by authorized administrators.

21.8 Analytics Categories

The platform shall provide analytics for:

- Sales
- Finance
- Projects
- AI
- Clients

- Documents
- Security
- Users
- Notifications
- Operational Performance

Future analytics categories shall be supported without requiring architectural redesign.

21.9 Reporting

Authorized users shall be able to:

- Generate reports.
- Filter reports.
- Export reports.
- Schedule reports.
- Share reports where authorized.
- Archive reports.
- Reports shall respect all platform authorization rules.

21.10 Historical Analysis

The platform shall support historical trend analysis.

Historical reporting shall support:

- Daily Trends
- Weekly Trends
- Monthly Trends
- Quarterly Trends
- Annual Trends
- Custom Time Periods

Historical analytics shall support business forecasting and operational optimization.

21.11 Automation Requirements

The platform shall automatically:

- Update dashboards.
- Refresh KPIs.
- Generate scheduled reports.
- Record analytics history.
- Calculate performance metrics.

- Monitor AI performance.
- Trigger alerts for predefined thresholds where configured.

Manual report generation shall remain available for authorized users.

21.12 Security Requirements

The Analytics & Business Intelligence module shall:

- Respect Organization isolation.
- Restrict dashboard visibility according to permissions.
- Protect confidential business information.
- Prevent unauthorized report access.
- Record report generation within the audit log.
- Protect exported data according to platform security policies.

21.13 Business Rules

Rule	Business Rule
BR-006	Organization Isolation
BR-007	Data Ownership
BR-020	Audit Logging
BR-022	Event-Driven Notifications

21.14 Acceptance Criteria

The module shall be considered complete when:

- Dashboards display accurate information.
- KPIs are calculated correctly.
- Reports are generated successfully.
- Historical analytics function correctly.
- Dashboard permissions are enforced.
- Scheduled reporting operates correctly.
- Analytics data remains consistent across the platform.

21.15 Out of Scope

The following functionality is excluded from

Version 1 implementation:

- Predictive analytics using machine learning.

- External Business Intelligence platform integrations.
- Real-time streaming analytics.
- Third-party analytics providers not selected by IO SKY.

The analytics architecture shall remain extensible for future analytical capabilities.

21.16 Developer Notes

This module defines the business requirements for analytics, reporting, dashboards, and business intelligence.

The development team is responsible for implementing dashboard infrastructure, KPI calculation, reporting services, historical analytics, authorization controls, and data visualization while ensuring compliance with all platform-wide business rules, security requirements, and audit obligations.

21.17 KPI Matrix

The platform shall maintain a centralized KPI

Matrix defining all Key Performance Indicators available within the IO SKY Platform.

Each KPI shall define:

- KPI Identifier
- KPI Name
- Business Purpose
- Formula or Calculation Method
- Data Sources
- Update Frequency
- Responsible Business Owner
- Applicable Dashboard(s)
- Authorized Roles
- Threshold Values
- Examples include, but are not limited to:
- Monthly Recurring Revenue (MRR)
- Annual Recurring Revenue (ARR)
- Active Clients
- Active Projects
- AI Scan Conversion Rate
- AI Agent Success Rate
- Customer Satisfaction Score
- Project Completion Rate

- Workflow Automation Rate
- Average Response Time
- Platform Availability

The KPI Matrix shall remain fully configurable by authorized administrators.

21.18 Dashboard Matrix

The platform shall maintain a standardized

Dashboard Matrix defining dashboard availability for each user role.

Each dashboard shall define:

- Dashboard Name
- Intended User Role(s)
- Available Widgets
- Accessible Business Data
- Refresh Frequency
- Export Permissions
- Drill-Down Capabilities
- Dashboard visibility shall always respect
- Organization isolation, user permissions, and business rules.

The dashboard architecture shall support the addition of future dashboards without requiring architectural redesign.

21.19 Report Catalog

The platform shall maintain a centralized catalog of standard reports.

Each report shall define:

- Report Identifier
- Report Name
- Business Purpose
- Data Sources
- Available Filters
- Supported Export Formats
- Scheduling Availability
- Authorized Roles
- Retention Policy
- Reports may include, but are not limited to:
- Executive Summary

- Revenue Report
- Sales Performance
- AI Scan Report
- AI Agent Performance
- Client Activity
- Project Performance
- Operational Performance
- Security Overview
- Compliance Overview
- The Report Catalog shall remain extensible.

21.20 Alert Threshold Matrix

The platform shall support configurable alert thresholds for operational monitoring.

Thresholds may be configured for:

- Revenue Targets
- AI Performance
- Failed Workflows
- System Availability
- Security Events
- Payment Failures
- Project Delays
- Customer Satisfaction
- Support Performance
- Platform Errors

When a configured threshold is exceeded, the platform shall automatically:

- Generate an alert.
- Notify authorized users according to the
- Notification Matrix.
- Record the event within the Audit Log.
- Trigger predefined workflows where applicable.

Threshold values shall remain configurable by authorized administrators.

21.21 Business Intelligence Data

Governance

The Analytics & Business Intelligence module shall ensure that all analytical information complies with platform-wide governance policies.

The platform shall:

- Respect Organization Isolation.
- Enforce Role-Based Access Control.
- Prevent unauthorized access to analytical data.
- Maintain data consistency across dashboards and reports.
- Record analytical data generation within the
- Audit Log.
- Support data validation before publication.

Business intelligence shall present only authorized and validated information.

21.22 Data Visualization Standards

The platform shall provide a consistent visualization framework for all dashboards and reports.

Visualization standards shall support:

- KPI Cards
- Tables
- Charts
- Trend Analysis
- Status Indicators
- Progress Indicators
- Comparative Analysis
- Time-Based Analysis

Visualizations shall remain responsive, accessible, and consistent throughout the platform.

21.23 Executive Decision Support

The Analytics & Business Intelligence module shall support executive decision-making by providing consolidated operational insights.

Executive dashboards may include:

- Business Growth
- Revenue Performance
- Operational Efficiency
- AI Utilization
- Customer Growth
- Project Portfolio Health
- Financial Performance
- Security Overview

- Platform Health
- Strategic Performance Indicators

Executive insights shall support informed business decisions while remaining configurable by authorized administrators.

21.24 Analytics Data Lifecycle

The platform shall manage analytical information throughout its lifecycle.

The lifecycle shall support:

- Data Collection
- Data Validation
- Data Processing
- KPI Calculation
- Dashboard Publication
- Historical Archiving
- Data Retention
- Secure Deletion where permitted

Historical analytical information shall remain available according to configurable retention policies.

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MODULE 16

22. Module 16 — Integration & API Management

Version	1.1	Priority	Critical
Status	Release Candidate	Business Owner	IO SKY

22.1 Purpose

The Integration & API Management module provides standardized connectivity between the

IO SKY Platform and external systems, applications, services, and business platforms.

The module enables secure, scalable, and governed integrations while ensuring interoperability, automation, and consistent data exchange across the platform ecosystem.

This module shall function as a shared platform service supporting all business modules.

22.2 Objectives

The Integration & API Management module shall enable the platform to:

- Connect external systems.
- Exchange business data securely.
- Automate cross-platform workflows.
- Manage API integrations.
- Support webhook communication.
- Synchronize business information.
- Monitor integration health.
- Maintain integration security.
- Support future integrations.
- Enable scalable interoperability.

22.3 Scope

The module includes:

- API Management
- Integration Registry
- Webhook Management

- Data Synchronization
- Integration Monitoring
- API Authentication
- API Authorization
- Integration Logging
- API Usage Monitoring
- Integration Configuration
- External Service Management
- Integration Reporting

22.4 Actors

The following actors interact with this module:

- Super Admin
- Admin (where authorized)
- Technical Operator (where authorized)
- AI Agents (where authorized)
- External Systems

Clients and Developers shall not directly manage platform integrations unless explicitly authorized.

22.5 API Management

The platform shall support standardized APIs for secure communication between internal and external systems.

API management shall support:

- API Registration
- API Version Management
- Authentication
- Authorization
- Usage Monitoring
- Error Handling
- Logging
- Availability Monitoring

22.6 Integration Registry

The platform shall maintain a centralized registry of all configured integrations.

Each integration shall define:

- Integration Identifier
- Name
- Description
- Connected System
- Integration Type
- Authentication Method
- Operational Status
- Business Owner
- Last Synchronization
- Version

22.7 Webhook Management

The platform shall support secure webhook communication.

Webhook management shall support:

- Outgoing Webhooks
- Incoming Webhooks
- Authentication
- Retry Policies
- Delivery Monitoring
- Failure Handling
- Logging
- Webhook processing shall comply with platform security policies.

22.8 Data Synchronization

The platform shall support controlled synchronization of business data between connected systems.

Synchronization shall support:

- Scheduled Synchronization
- Event-Driven Synchronization
- Manual Synchronization
- Incremental Updates
- Conflict Detection
- Synchronization History
- Error Recovery

22.9 Integration Monitoring

The platform shall continuously monitor integration health.

Monitoring shall include:

- Availability
- Response Status
- Synchronization Status
- Failed Requests
- Retry Attempts
- Performance Metrics
- Integration Errors
- Monitoring results shall be available to authorized users.

22.10 Automation Requirements

The platform shall automatically:

- Execute integration workflows.
- Synchronize configured data.
- Retry failed operations where applicable.
- Record integration history.
- Generate integration alerts.
- Notify authorized users of critical failures.
- Maintain integration logs.

22.11 Security Requirements

The Integration & API Management module shall:

- Protect API credentials.
- Enforce authentication.
- Enforce authorization.
- Record integration activities.
- Prevent unauthorized access.
- Support encrypted communication.
- Protect confidential business data.

22.12 Business Rules

Rule	Business Rule
BR-006	Organization Isolation
BR-007	Data Ownership
BR-020	Audit Logging

BR-022	Event-Driven Notifications

22.13 Acceptance Criteria

The module shall be considered complete when:

- APIs operate securely.
- Integrations function correctly.
- Synchronization completes successfully.
- Integration monitoring is operational.
- Security controls are enforced.
- Integration logs are maintained.
- Alerts are generated where required.

22.14 Out of Scope

The following functionality is excluded from

Version 1 implementation:

- Proprietary integrations not approved by IO
- SKY.
- Unsupported third-party platforms.
- Legacy system migration services.

The architecture shall remain extensible to support future integrations.

22.15 Developer Notes

This module defines the business requirements for API management, integrations, synchronization, and interoperability.

The development team is responsible for implementing secure APIs, integration services, webhook processing, synchronization mechanisms, monitoring capabilities, and authorization controls while complying with all platform-wide business rules.

22.16 Integration Catalog

The platform shall maintain a centralized catalog of all supported integrations.

The catalog shall define:

- Integration Name
- Business Purpose
- Connected Platform

- Data Exchanged
- Authentication Method
- Synchronization Type
- Supported Operations
- Status
- Business Owner

The Integration Catalog shall support the addition of future integrations without requiring architectural redesign.

22.17 API Lifecycle Management

The platform shall maintain a standardized lifecycle for all APIs exposed or consumed by the IO SKY Platform.

The API lifecycle shall support:

- API Registration
- API Development
- Testing
- Approval
- Publication
- Version Management
- Deprecation
- Retirement

Each API shall maintain version history and operational status throughout its lifecycle.

22.18 API Security Policies

The platform shall enforce centralized security policies governing all API communications.

API security policies shall include:

- Authentication Requirements
- Authorization Requirements
- Credential Management
- Token Management
- Secure Communication
- Request Validation
- Response Validation
- Access Restrictions
- Audit Logging

- Security Monitoring

All API communications shall comply with platform-wide security policies.

22.19 Integration Failure & Recovery

The platform shall detect, record, and recover from integration failures where technically feasible.

Failure management shall support:

- Failure Detection
- Automatic Retry
- Retry Limits
- Manual Recovery
- Error Logging
- Failure Notifications
- Recovery Status
- Recovery History

Critical integration failures shall notify authorized users according to the Notification

Matrix.

22.20 Data Mapping & Transformation

The platform shall support standardized mapping and transformation of data exchanged between integrated systems.

Data transformation shall support:

- Field Mapping
- Data Validation
- Format Conversion
- Data Normalization
- Default Values
- Optional Fields
- Required Fields
- Transformation Rules

Transformation logic shall preserve data integrity throughout all integration processes.

22.21 API Usage Governance

The platform shall monitor and govern API usage to ensure reliable and controlled platform operations.

Governance capabilities shall include:

- Usage Monitoring
- Request Limits
- Rate Limiting
- Usage Statistics
- Error Tracking
- Performance Monitoring
- Quota Management
- Capacity Monitoring

Usage policies shall remain configurable by authorized administrators.

22.22 Integration Version Management

The platform shall support version management for all configured integrations.

Version management shall support:

- Version Registration
- Compatibility Tracking
- Upgrade Management
- Deprecation Tracking
- Rollback Support
- Version History
- Release Status

Integration version history shall remain available for audit and operational purposes.

22.23 Supported Integration Matrix

The platform shall maintain a centralized

Supported Integration Matrix documenting all approved integrations.

Each integration shall define:

- Integration Name
- Connected Platform
- Business Purpose
- Supported Data
- Authentication Method
- Synchronization Type
- Operational Status
- Supported Features
- Business Owner

- The matrix shall remain extensible to support future integrations.

22.24 Integration Dashboard & Monitoring

The platform shall provide centralized dashboards for monitoring integration operations.

Authorized users shall be able to monitor:

- Active Integrations
- Integration Status
- API Availability
- Synchronization Status
- Failed Requests
- Retry Operations
- Processing Times
- Integration Performance
- Error History
- Operational Health

Dashboard visibility shall comply with platform authorization policies.

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MODULE 17

23. Module 17 — Workflow & Business Process Management

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

23.1 Purpose

The Workflow & Business Process Management module provides centralized orchestration of business processes across the IO SKY Platform.

The module enables standardized, automated, and controlled execution of operational workflows while ensuring consistency, transparency, governance, and scalability throughout the platform.

This module shall function as a shared platform service supporting all business modules.

23.2 Objectives

The Workflow & Business Process Management module shall enable the platform to:

- Standardize business processes.
- Automate operational workflows.
- Coordinate cross-module activities.
- Support approval workflows.
- Manage workflow execution.
- Monitor workflow performance.
- Maintain workflow history.
- Support human intervention where required.
- Improve operational efficiency.
- Enable scalable business automation.

23.3 Scope

The module includes:

- Workflow Management
- Workflow Templates

- Workflow Execution
- Business Process Automation
- Approval Workflows
- Task Orchestration
- Event-Driven Workflows
- Scheduled Workflows
- Workflow Monitoring
- Workflow Reporting
- Workflow History
- Workflow Configuration

23.4 Actors

The following actors interact with this module:

- Super Admin
- Admin
- Technical Operator (where authorized)
- AI Agents (where authorized)
- System Services

Clients and Developers shall only interact with workflows where explicitly authorized by platform business rules.

23.5 Workflow Management

The platform shall support centralized management of business workflows.

Each workflow shall define:

- Workflow Identifier
- Workflow Name
- Business Purpose
- Trigger Event
- Workflow Steps
- Responsible Roles
- Required Approvals
- Expected Outcomes
- Status
- Version

Workflow definitions shall remain configurable by authorized administrators.

23.6 Workflow Templates

The platform shall support reusable workflow templates.

Templates may be created for:

- AI Scan Processing
- Lead Qualification
- Client Onboarding
- Project Delivery
- Invoice Processing
- Subscription Management
- Document Approval
- AI Agent Escalation
- Support Requests
- Security Incidents
- Workflow templates shall support future business processes.

23.7 Workflow Execution

The platform shall execute workflows automatically or manually according to business rules.

Workflow execution shall support:

- Sequential Steps
- Parallel Steps
- Conditional Logic
- Approval Gates
- Human Intervention
- AI Execution
- Scheduled Execution
- Event-Based Execution
- Each workflow execution shall be recorded within the Audit Log.

23.8 Approval Workflows

The platform shall support configurable approval workflows.

Approval processes may include:

- AI Scan Approval
- Proposal Approval
- Invoice Approval
- Document Approval

- Project Approval
- User Access Approval
- Configuration Approval
- Administrative Approval

Approval requirements shall remain configurable by authorized administrators.

23.9 Task Orchestration

The platform shall coordinate tasks between users, AI Agents, and system services.

Task orchestration shall support:

- Task Assignment
- Task Prioritization
- Task Dependencies
- Task Completion
- Escalation
- Notifications
- Deadlines
- Workflow Continuation

23.10 Event-Driven Workflows

The platform shall support workflows initiated by business events.

Examples include:

- User Registration
- Payment Confirmation
- AI Scan Completion
- Opportunity Won
- Project Completion
- Subscription Renewal
- Security Incident
- Document Approval

Business events shall automatically initiate configured workflows where applicable.

23.11 Scheduled Workflows

The platform shall support scheduled execution of recurring workflows.

Scheduled workflows may include:

- Daily Reports

- Subscription Renewals
- Data Synchronization
- AI Performance Reviews
- Maintenance Tasks
- Compliance Checks
- Notification Processing
- System Health Reviews

23.12 Automation Requirements

The platform shall automatically:

- Execute configured workflows.
- Route workflow tasks.
- Trigger notifications.
- Maintain workflow history.
- Detect workflow failures.
- Retry supported workflow actions.
- Generate workflow metrics.
- Escalate workflow exceptions where required.

23.13 Security Requirements

The Workflow & Business Process Management module shall:

- Enforce Role-Based Access Control.
- Respect Organization Isolation.
- Record workflow execution.
- Prevent unauthorized workflow modifications.
- Protect workflow configurations.
- Maintain complete audit history.

23.14 Business Rules

Rule	Business Rule
BR-006	Organization Isolation
BR-007	Data Ownership
BR-020	Audit Logging
BR-022	Event-Driven Notifications

23.15 Acceptance Criteria

The module shall be considered complete when:

- Workflows execute correctly.
- Workflow history is maintained.
- Approval workflows function correctly.
- Event-driven workflows trigger successfully.
- Scheduled workflows execute as configured.
- Workflow security is enforced.
- Workflow monitoring operates successfully.

23.16 Out of Scope

The following functionality is excluded from

Version 1 implementation:

- External workflow engines not approved by
- IO SKY.
- Business Process Mining platforms.
- Third-party workflow orchestration platforms.

The architecture shall remain extensible to support future workflow technologies.

23.17 Developer Notes

This module defines the business requirements for workflow orchestration, business process automation, approval management, task coordination, and workflow monitoring.

The development team is responsible for implementing the workflow engine, execution services, approval mechanisms, monitoring capabilities, automation infrastructure, and authorization controls while ensuring compliance with all platform-wide business rules.

23.18 Workflow Lifecycle

Management

The platform shall maintain a standardized lifecycle for all business workflows.

The workflow lifecycle shall support:

- Workflow Creation
- Draft Management
- Validation
- Approval

- Publication
- Activation
- Modification
- Version Updates
- Archiving
- Retirement
- Each workflow shall maintain a complete lifecycle history.

23.19 Workflow State

Management

The platform shall maintain standardized workflow states throughout workflow execution.

The default workflow states shall include:

- Draft
- Pending Approval
- Approved
- Active
- Running
- Waiting for Input
- Waiting for Approval
- Suspended
- Completed
- Failed
- Cancelled
- Archived
- Workflow state transitions shall follow predefined business rules.

23.20 Workflow Exception & Recovery

The platform shall support standardized handling of workflow exceptions.

Exception management shall support:

- Failure Detection
- Error Classification
- Automatic Retry
- Retry Limits
- Manual Intervention
- Workflow Recovery

- Escalation
- Error Logging
- Recovery History

Critical workflow failures shall notify authorized users according to the Notification Matrix.

23.21 Workflow Template Catalog

The platform shall maintain a centralized catalog of reusable workflow templates.

The catalog may include, but is not limited to:

- Lead Qualification
- Client Onboarding
- AI Scan Processing
- Discovery Call Follow-up
- Proposal Approval
- Project Delivery
- Invoice Approval
- Subscription Renewal
- Support Request Handling
- AI Agent Escalation
- Security Incident Response

The Workflow Template Catalog shall remain extensible for future business processes.

23.22 Workflow Metrics & Performance

The platform shall continuously monitor workflow performance.

Workflow metrics shall include:

- Total Executions
- Successful Executions
- Failed Executions
- Average Completion Time
- Waiting Time
- Approval Time
- Automation Rate
- Human Intervention Rate
- AI Execution Rate
- Bottleneck Identification

Performance information shall support operational optimization and business reporting.

23.23 Workflow Version Management

The platform shall support version management for workflow definitions.

Version management shall support:

- Version Registration
- Version History
- Version Comparison
- Approval Before Publication
- Rollback Support
- Change Tracking
- Release Status
- Workflow version history shall remain available for audit purposes.

23.24 Workflow Dependency Matrix

The platform shall maintain a Workflow

Dependency Matrix documenting relationships between workflows.

Each dependency shall define:

- Source Workflow
- Dependent Workflow
- Trigger Event
- Required Conditions
- Responsible Module
- Expected Outcome

Workflow dependencies shall prevent inconsistent execution across business processes.

23.25 Workflow Governance & Compliance

The platform shall enforce governance policies for all workflow operations.

Workflow governance shall include:

- Workflow Ownership
- Role-Based Authorization
- Approval Requirements
- Audit Logging
- Business Rule Validation
- Compliance Monitoring
- Workflow Documentation

- Operational Accountability

All workflow activities shall comply with platform-wide governance, security, and compliance policies.

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MODULE 18

24. Module 18 — Platform Configuration & System Administration

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

24.1 Purpose

The Platform Configuration & System

Administration module provides centralized configuration, administration, and operational control of the IO SKY Platform.

The module enables authorized administrators to configure business settings, platform behavior, operational policies, and system-wide functionality without requiring software modifications.

This module shall function as a shared platform service supporting all platform modules.

24.2 Objectives

The Platform Configuration & System

Administration module shall enable the platform to:

- Configure global platform settings.
- Manage business configuration.
- Maintain operational policies.
- Configure AI behavior.
- Configure workflow defaults.
- Manage feature availability.
- Configure notifications.
- Configure integrations.
- Configure security settings.
- Centralize administrative management.

24.3 Scope

The module includes:

- Global Configuration
- Organization Settings
- System Administration
- Platform Configuration
- Feature Management
- AI Configuration
- Workflow Configuration
- Notification Configuration
- Security Configuration
- Business Settings
- Regional Settings
- Operational Policies

24.4 Actors

The following actors interact with this module:

- Super Admin
- Admin (where authorized)
- Technical Operator (where explicitly authorized)

Clients, Developers, and AI Agents shall not modify platform configuration unless explicitly authorized.

24.5 Global Platform Configuration

The platform shall support centralized management of global platform settings.

Global settings may include:

- Platform Name
- Branding
- Company Information
- Default Language
- Regional Settings
- Date & Time Configuration
- Currency
- Time Zone
- Platform Availability
- Maintenance Mode

Global settings shall apply consistently across all platform modules.

24.6 Organization Configuration

The platform shall support configuration of organization-specific settings.

Organization settings may include:

- Organization Profile
- Business Information
- Branding
- Default Preferences
- Notification Preferences
- AI Preferences
- Business Rules
- Regional Settings
- Organization settings shall remain isolated per organization.

24.7 Feature Management

The platform shall support controlled activation and deactivation of platform features.

Feature management shall support:

- Feature Availability
- Feature Visibility
- Module Availability
- Experimental Features
- Future Features
- Organization-Specific Features

Feature availability shall remain configurable by authorized administrators.

24.8 AI Configuration

The platform shall support centralized AI configuration.

AI configuration may include:

- Default AI Provider
- AI Models
- AI Permissions
- AI Knowledge Sources
- AI Safety Policies
- AI Response Settings

- AI Usage Policies
- AI Operational Limits

24.9 Workflow Configuration

Authorized administrators shall be able to configure:

- Workflow Templates
- Approval Policies
- Workflow Defaults
- Escalation Policies
- Automation Rules
- Scheduled Workflows
- Workflow Notifications

24.10 Notification Configuration

The platform shall support centralized notification management.

Configuration shall include:

- Email Templates
- Notification Channels
- Notification Preferences
- Delivery Policies
- Escalation Rules
- Reminder Policies

24.11 Security Configuration

Authorized administrators shall be able to configure:

- Authentication Policies
- Password Policies
- Session Policies
- Multi-Factor Authentication
- Access Policies
- Security Alerts
- AI Security Policies

24.12 Automation Requirements

The platform shall automatically:

- Apply approved configuration changes.
- Validate configuration integrity.
- Record configuration changes.
- Notify authorized users of critical configuration changes.
- Maintain configuration history.

24.13 Security Requirements

The Platform Configuration & System

Administration module shall:

- Restrict configuration access.
- Record administrative activities.
- Protect configuration integrity.
- Maintain configuration history.
- Enforce authorization policies.
- Support audit requirements.

24.14 Business Rules

Rule	Business Rule
BR-006	Organization Isolation
BR-007	Data Ownership
BR-020	Audit Logging
BR-022	Event-Driven Notifications

24.15 Acceptance Criteria

The module shall be considered complete when:

- Configuration changes are applied correctly.
- Administrative permissions are enforced.
- Configuration history is maintained.
- Platform settings remain consistent.
- Security policies are configurable.
- AI configuration functions correctly.

24.16 Out of Scope

The following functionality is excluded from

Version 1 implementation:

- Source code editing.
- Infrastructure configuration outside the IO
- SKY Platform.
- Third-party platform administration not managed by IO SKY.

The architecture shall remain extensible for future administrative capabilities.

24.17 Developer Notes

This module defines the business requirements for platform configuration, administrative management, operational policies, and centralized system administration.

The development team is responsible for implementing configuration management, administrative interfaces, authorization controls, configuration validation, and auditing while ensuring compliance with all platform-wide business rules.

24.18 Configuration Lifecycle

Management

The platform shall maintain a standardized lifecycle for all platform configurations.

The configuration lifecycle shall support:

- Configuration Creation
- Draft Configuration
- Validation
- Approval
- Publication
- Activation
- Modification
- Version Updates
- Archiving
- Retirement
- Each configuration shall maintain a complete lifecycle history.

24.19 Configuration Version

Management

The platform shall support version management for all platform configurations.

Version management shall support:

- Version Registration
- Version History
- Change Tracking
- Configuration Comparison
- Rollback Support
- Release Approval
- Active Version Identification

Historical configuration versions shall remain available for audit purposes.

24.20 Configuration Templates

The platform shall maintain reusable configuration templates.

Templates may include:

- Organization Configuration
- AI Configuration
- Workflow Configuration
- Notification Configuration
- Security Configuration
- Regional Configuration
- Branding Configuration
- Feature Configuration

Configuration templates shall simplify platform administration while ensuring consistency.

24.21 Configuration Validation

The platform shall validate configuration changes before publication.

Validation shall include:

- Required Fields
- Dependency Validation
- Permission Validation
- Business Rule Validation
- Security Validation
- Data Integrity Validation
- Configuration Consistency

Invalid configurations shall not become active until validation requirements have been satisfied.

24.22 Feature Flag Management

The platform shall support centralized management of feature availability.

Feature flag management shall support:

- Global Feature Flags
- Organization-Specific Features
- User Role Features
- Experimental Features
- Beta Features
- Scheduled Activation
- Scheduled Deactivation
- Feature Rollback

Feature availability shall remain configurable without requiring software modifications.

24.23 Configuration Backup & Restore

The platform shall support secure backup and restoration of platform configurations.

Configuration management shall support:

- Configuration Backups
- Backup History
- Restore Operations
- Version Recovery
- Configuration Export
- Configuration Import
- Validation After Restore
- Restoration activities shall be recorded within the Audit Log.

24.24 Configuration Dashboard

Authorized administrators shall have access to centralized configuration dashboards.

The dashboard shall provide visibility into:

- Active Configurations
- Configuration Versions
- Pending Changes
- Recent Changes
- Configuration Health
- Validation Status

- Failed Validations
- Feature Availability
- Administrative Activity

Dashboard visibility shall comply with platform authorization policies.

24.25 Configuration Governance & Compliance

The platform shall enforce governance policies for all platform configurations.

Configuration governance shall include:

- Configuration Ownership
- Approval Requirements
- Role-Based Authorization
- Audit Logging
- Business Rule Validation
- Security Compliance
- Change Accountability
- Configuration Documentation

All configuration activities shall comply with platform-wide governance, security, and compliance policies.

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MODULE 19

25. Module 19 — Platform Operations & Maintenance

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

25.1 Purpose

The Platform Operations & Maintenance module defines the operational management requirements for maintaining, monitoring, updating, and supporting the IO SKY Platform throughout its operational lifecycle.

The module ensures platform stability, availability, maintainability, recoverability, and operational continuity.

This module shall function as a shared platform service supporting all business modules.

25.2 Objectives

The Platform Operations & Maintenance module shall enable the platform to:

- Maintain operational availability.
- Support scheduled maintenance.
- Manage platform updates.
- Monitor platform health.
- Support backup and recovery.
- Maintain operational continuity.
- Support incident response.
- Support release management.
- Maintain operational documentation.
- Ensure long-term platform reliability.

25.3 Scope

The module includes:

- Platform Operations
- Maintenance Management
- Platform Monitoring

- Release Management
- Backup Management
- Recovery Management
- Operational Health
- Service Availability
- Incident Response
- Maintenance Scheduling
- Capacity Monitoring
- Operational Reporting

25.4 Actors

The following actors interact with this module:

- Super Admin
- Technical Operator
- Authorized Admins
- System Services

Clients, Developers, and AI Agents shall not manage platform operations unless explicitly authorized.

25.5 Platform Monitoring

The platform shall continuously monitor operational health.

Monitoring shall include:

- Platform Availability
- Service Status
- Database Health
- Integration Health
- AI Service Health
- Background Jobs
- Queue Status
- Storage Health
- Network Connectivity
- System Performance

25.6 Maintenance Management

The platform shall support controlled maintenance activities.

Maintenance management shall support:

- Planned Maintenance
- Emergency Maintenance
- Maintenance Scheduling
- Maintenance Notifications
- Maintenance Windows
- Maintenance History

25.7 Release Management

The platform shall support controlled software releases.

Release management shall support:

- Release Planning
- Release Approval
- Deployment Tracking
- Release History
- Rollback Support
- Version Tracking

25.8 Backup & Recovery

The platform shall support secure backup and recovery procedures.

Backup management shall support:

- Scheduled Backups
- Manual Backups
- Backup Verification
- Recovery Procedures
- Recovery Validation
- Backup History

25.9 Capacity Management

The platform shall monitor operational capacity.

Capacity monitoring shall include:

- Storage Utilization
- Compute Resources
- AI Consumption
- API Utilization
- Database Capacity

- Integration Capacity
- Queue Capacity

25.10 Incident Response

The platform shall support operational incident management.

Incident management shall support:

- Incident Detection
- Incident Registration
- Incident Classification
- Assignment
- Resolution
- Root Cause Documentation
- Incident Closure

25.11 Automation Requirements

The platform shall automatically:

- Monitor platform health.
- Detect operational failures.
- Generate alerts.
- Maintain monitoring history.
- Record operational events.
- Trigger backup procedures where configured.
- Notify authorized personnel of critical incidents.

25.12 Security Requirements

The Platform Operations & Maintenance module shall:

- Protect operational information.
- Restrict operational administration.
- Record maintenance activities.
- Protect backup integrity.
- Protect monitoring information.
- Maintain audit history.

25.13 Business Rules

Rule	Business Rule

BR-006	Organization Isolation
BR-007	Data Ownership
BR-020	Audit Logging
BR-022	Event-Driven Notifications

25.14 Acceptance Criteria

The module shall be considered complete when:

- Platform monitoring operates continuously.
- Maintenance procedures function correctly.
- Backup and recovery processes are available.
- Operational incidents are managed successfully.
- Release management is operational.
- Capacity monitoring functions correctly.

25.15 Out of Scope

The following functionality is excluded from

Version 1 implementation:

- Third-party infrastructure management outside the IO SKY Platform.
- External cloud provider administration.
- Hardware lifecycle management.

The architecture shall remain extensible for future operational capabilities.

25.16 Developer Notes

This module defines the business requirements for platform operations, maintenance, monitoring, release management, backup, recovery, and operational continuity.

The development team is responsible for implementing monitoring services, operational tooling, maintenance workflows, release mechanisms, backup services, recovery capabilities, and operational dashboards while ensuring compliance with all platform-wide business rules.

25.17 Operations Lifecycle

Management

The platform shall maintain a standardized lifecycle for operational services and maintenance activities.

The operations lifecycle shall support:

- Planning
- Scheduling
- Preparation
- Execution
- Validation
- Completion
- Review
- Continuous Improvement
- Archiving

All operational activities shall maintain a complete lifecycle history.

25.18 Maintenance Policy

Management

The platform shall support centralized management of maintenance policies.

Maintenance policies shall define:

- Planned Maintenance
- Emergency Maintenance
- Maintenance Windows
- Notification Requirements
- Approval Requirements
- Operational Impact
- Maintenance Priority
- Recovery Validation

Maintenance policies shall remain configurable by authorized administrators.

25.19 Release & Deployment

Governance

The platform shall maintain governance over software releases and deployments.

Release governance shall support:

- Release Planning
- Change Approval
- Release Scheduling
- Deployment Validation
- Rollback Procedures
- Release Documentation

- Version Tracking
- Deployment History

All releases shall comply with platform governance and audit requirements.

25.20 Backup Retention & Recovery Policies

The platform shall maintain standardized policies governing backup retention and recovery.

Policies shall define:

- Backup Frequency
- Retention Periods
- Backup Verification
- Recovery Objectives
- Recovery Validation
- Archive Policies
- Secure Storage
- Recovery Documentation

Backup policies shall support operational continuity and disaster recovery.

25.21 Operational Dashboard

Authorized administrators shall have access to centralized operational dashboards.

Operational dashboards shall provide visibility into:

- Platform Availability
- Active Incidents
- Scheduled Maintenance
- Backup Status
- Release Status
- System Health
- Capacity Utilization
- Integration Health
- AI Service Status
- Operational Alerts

Dashboard visibility shall comply with platform authorization policies.

25.22 Service Level Management

The platform shall support monitoring of operational service levels.

Service level management shall include:

- Availability Targets
- Response Targets
- Recovery Targets
- Operational Performance
- Incident Resolution Performance
- Maintenance Performance
- Capacity Performance
- Service History

Service level information shall support operational reporting and continuous improvement.

25.23 Operational Metrics & Reporting

Performance

The platform shall continuously monitor operational performance.

Operational metrics shall include:

- Platform Uptime
- Incident Frequency
- Recovery Time
- Maintenance Duration
- Release Success Rate
- Backup Success Rate
- System Availability
- Capacity Utilization
- Integration Availability
- AI Service Availability

Performance information shall support proactive operational management.

25.24 Operations Governance & Compliance

The platform shall enforce governance policies for all operational activities.

Operational governance shall include:

- Operational Ownership
- Role-Based Authorization
- Maintenance Approval
- Audit Logging
- Change Accountability
- Compliance Monitoring

- Operational Documentation
- Continuous Improvement

All operational activities shall comply with platform-wide governance, security, and compliance policies.

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MODULE 20

26. Module 20 — Global Non-Functional Requirements

Version	1.1	**Priority**	Critical
Status	Release Candidate	**Business Owner**	IO SKY

26.1 Purpose

This module defines the global non-functional requirements applicable to the entire IO SKY

Platform.

These requirements establish the expected quality attributes, operational standards, and performance characteristics of the platform independently of specific business functionality.

All platform modules shall comply with these requirements.

26.2 Objectives

The Global Non-Functional Requirements shall ensure:

- Platform reliability.
- High availability.
- Performance.
- Scalability.
- Security.
- Maintainability.
- Usability.
- Accessibility.
- Recoverability.
- Operational excellence.

26.3 Scope

These requirements apply to:

- Public Website
- AI Scan Platform
- Client Portal

- Developer Portal
- Admin Portal
- Super Admin Portal
- CRM
- AI Services
- Workflow Engine
- Notification Services
- File Management
- Integrations
- Analytics
- Platform Infrastructure

26.4 Performance Requirements

The platform shall provide responsive performance under normal operational conditions.

Performance requirements include:

- Responsive user interactions.
- Efficient data processing.
- Optimized workflow execution.
- Efficient report generation.
- Efficient AI processing.
- Background task execution.
- Optimized database operations.
- Performance shall remain consistent across supported environments.

26.5 Availability Requirements

The platform shall be designed to maximize operational availability.

Availability shall include:

- Continuous service availability.
- Planned maintenance support.
- Operational monitoring.
- Service recovery.
- Redundancy where applicable.
- Failure detection.

26.6 Scalability Requirements

The platform architecture shall support future growth.

Scalability shall support:

- Increasing users.
- Organizations.
- AI workloads.
- Data volumes.
- Workflow executions.
- Integrations.
- Platform services.

The architecture shall allow horizontal and vertical expansion where appropriate.

26.7 Reliability Requirements

The platform shall operate consistently under expected workloads.

Reliability shall include:

- Stable operation.
- Error handling.
- Transaction integrity.
- Consistent processing.
- Service continuity.
- Controlled failure recovery.

26.8 Security Requirements

The platform shall comply with all platform-wide security policies.

Security shall include:

- Authentication.
- Authorization.
- Encryption.
- Audit Logging.
- Secure Communication.
- Data Protection.
- Access Control.
- AI Security.
- Security Monitoring.

26.9 Maintainability Requirements

The platform shall support efficient long-term maintenance.

Maintainability shall include:

- Modular architecture.
- Clear separation of responsibilities.
- Configuration management.
- Documentation.
- Version management.
- Operational monitoring.

26.10 Usability Requirements

The platform shall provide a consistent user experience.

Usability shall include:

- Consistent navigation.
- Predictable workflows.
- Clear terminology.
- User feedback.
- Error messages.
- Responsive interfaces.
- Efficient task completion.

26.11 Accessibility Requirements

The platform shall support accessible user interaction wherever reasonably applicable.

Accessibility shall include:

- Keyboard navigation.
- Readable interfaces.
- Consistent layouts.
- Clear visual hierarchy.
- Accessible form interactions.
- Understandable validation messages.

26.12 Data Integrity Requirements

The platform shall protect the integrity of all business data.

Data integrity shall include:

- Validation.
- Consistency.
- Controlled modifications.

- Auditability.
- Recovery support.
- Transaction integrity.

26.13 Disaster Recovery

Requirements

The platform shall support recovery following operational failures.

Recovery capabilities shall include:

- Backup recovery.
- Service restoration.
- Data recovery.
- Recovery validation.
- Operational continuity.

26.14 Logging & Monitoring

Requirements

The platform shall maintain operational visibility.

Monitoring shall include:

- Platform Health.
- Security Events.
- Operational Events.
- Workflow Events.
- AI Operations.
- Integrations.
- Errors.
- Alerts.

26.15 Compliance Requirements

The platform shall support applicable legal and organizational compliance requirements.

Compliance shall include:

- Auditability.
- Data governance.
- Security governance.
- Operational governance.

- Privacy protection.
- Business accountability.

26.16 Privacy & GDPR

The platform shall be designed and operated in accordance with applicable data protection law, including the General Data Protection Regulation (GDPR) where applicable.

At the business level, the platform shall:

- Process personal data only for defined, legitimate business purposes.
- Collect and retain no more personal data than is necessary for those purposes.
- Provide data subjects with the ability to exercise their applicable legal rights.
- Maintain accountability for how personal data is processed, stored, and protected.
- Ensure that subprocessors and AI providers engaged by the platform are held to equivalent data protection standards.
- Notify affected parties and authorities of a data breach where required by law.

Detailed legal and technical implementation of these objectives — including lawful basis per processing activity, consent management, data subject request handling, retention schedules, breach-notification procedures, subprocessor and AI-provider data-handling terms, and specific security controls — is governed by the Security & Compliance Specification, which this Master SRS references as an authoritative companion document.

Where a requirement in this section and the Security & Compliance Specification appear to conflict, the Security & Compliance Specification governs for privacy and legal matters, and this Master SRS governs for platform functional behavior; any such conflict shall be escalated to the Product Owner rather than resolved silently.

26.17 Business Rules

Rule	Business Rule
This module shall comply with all global business rules defined within this Software Requirements Specification.	

26.18 Acceptance Criteria

The module shall be considered complete when:

- All platform modules comply with these requirements.
- Non-functional requirements are consistently enforced.
- Platform quality standards are maintained.
- Operational standards are satisfied.

- Security standards are satisfied.
- Reliability objectives are achieved.

26.19 Out of Scope

The following items are excluded from this specification:

- Vendor-specific implementation details.
- Technology-specific architecture decisions.
- Programming language requirements.
- Infrastructure implementation details.
- These decisions remain the responsibility of the development team.

26.20 Developer Notes

This module defines the global quality requirements applicable across the entire IO

SKY Platform.

The development team is responsible for implementing technical solutions that satisfy these non-functional requirements while selecting the most appropriate architecture, technologies, and implementation strategies.

26.21 Quality Attribute Governance

The platform shall maintain governance over all non-functional quality attributes.

Quality governance shall include:

- Quality Ownership
- Quality Reviews
- Change Approval
- Quality Validation
- Continuous Monitoring
- Compliance Verification
- Quality Documentation
- Improvement Planning
- Quality governance shall apply across all platform modules.

26.22 Performance & Capacity

Metrics

The platform shall continuously monitor global performance and capacity metrics.

Metrics shall include:

- Platform Response Time
- Transaction Throughput
- AI Processing Performance
- Workflow Performance
- API Performance
- Database Performance
- Resource Utilization
- Capacity Trends

Performance metrics shall support proactive optimization and operational reporting.

26.23 Availability & Resilience

Standards

The platform shall maintain standards supporting operational continuity.

Availability standards shall include:

- High Availability
- Failure Detection
- Automatic Recovery
- Service Redundancy
- Operational Resilience
- Recovery Validation
- Planned Maintenance Management
- Service Restoration
- The platform shall be designed to minimize service interruptions.

26.24 Scalability Governance

The platform shall support controlled scalability across all platform services.

Scalability governance shall include:

- User Growth
- Organization Growth
- AI Workload Growth
- Data Growth
- Integration Growth
- Workflow Growth
- Storage Growth

- Processing Capacity
- Scalability planning shall support future business expansion.

26.25 Security Assurance

The platform shall maintain continuous security assurance.

Security assurance shall include:

- Security Reviews
- Vulnerability Assessments
- Security Monitoring
- Incident Evaluation
- Access Reviews
- Policy Compliance
- Risk Assessments
- Security Improvement

Security assurance shall support continuous protection of platform operations.

26.26 Maintainability & Support

Standards

The platform shall support efficient long-term maintenance.

Maintainability standards shall include:

- Modular Design
- Configuration Management
- Version Management
- Operational Documentation
- Support Procedures
- Change Management
- Knowledge Transfer
- Continuous Maintenance
- These standards shall improve long-term platform sustainability.

26.27 Continuous Improvement

The platform shall support continuous improvement throughout its operational lifecycle.

Continuous improvement shall include:

- Performance Reviews

- User Feedback Evaluation
- Operational Reviews
- Security Reviews
- AI Performance Reviews
- Workflow Optimization
- Process Improvement
- Platform Evolution

Continuous improvement activities shall be documented where appropriate.

26.28 Global Governance & Compliance

The platform shall enforce governance and compliance across the entire IO SKY Platform.

Global governance shall include:

- Platform Ownership
- Business Accountability
- Role-Based Authorization
- Audit Logging
- Compliance Monitoring
- Risk Management
- Policy Enforcement
- Continuous Governance

All platform modules shall comply with these global governance requirements.

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APPENDIX A

Appendix A — Glossary of Terms

Requirement identifier prefixes used throughout this specification:

Prefix	Meaning
BR	Business Rule — a mandatory platform-wide requirement (see Appendix B).
PP	Product Principle — a guiding principle applied to every module (see Appendix C).

Definitions of terms used throughout this specification.

Term	Definition
AI Scan	An operational or cybersecurity assessment performed by the platform that produces an expert-reviewed report.
Organization	An isolated tenant that owns all of its business records; users belong to at least one Organization.
Client	An external customer user who accesses the Client Portal.
Developer	An internal delivery user who may access only projects explicitly assigned to them.
Admin	An internal user who manages customers, projects, and communication.
Super Admin	A platform-level administrator with full governance authority.
Technical Operator	A user with configurable technical permissions defined by the Super Admin.
Report Lifecycle	Draft → AI Processing → Expert Review → Approved → Published → Archived (optional).

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APPENDIX B

Appendix B — Business Rule Index

Index of the global Business Rules referenced throughout this specification.

ID	Business Rule
BR-001	Authentication Required
BR-002	Single Identity
BR-003	Automatic Portal Routing
BR-004	Invitation-Based Access
BR-005	Organization Ownership
BR-006	Organization Isolation
BR-007	Data Ownership
BR-008	AI Supports Human Decision-Making
BR-009	Human Approval Required
BR-010	AI Transparency
BR-011	Customer Communication
BR-012	Internal Communication
BR-013	Least Privilege
BR-014	Multi-Factor Authentication
BR-015	Session Security
BR-016	Report Lifecycle
BR-017	Version Control
BR-018	Project Ownership
BR-019	Approval Workflow
BR-020	Audit Logging
BR-021	Audit Visibility
BR-022	Event-Driven Notifications

BR-023	Payment Confirmation
BR-024	Technical Operator Access
BR-025	Technical Operator Accountability
BR-026	Workflow Automation

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APPENDIX C

Appendix C — Product Principle Index

Index of the Product Principles that apply to every module.

ID	Product Principle
PP-001	Security by Default
PP-002	Human Oversight
PP-003	Organization Isolation
PP-004	Single Source of Truth
PP-005	Role-Based Access Control
PP-006	Complete Auditability
PP-007	Enterprise First
PP-008	Consistency

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APPENDIX D

Appendix D — Open Product Owner Decisions

OPD-002 (Notification Retry & Escalation Policy) and OPD-004 (Privacy & GDPR architecture) have been resolved and incorporated (see §17.15 and §20.16 respectively, and the Security & Compliance Specification). The following items remain open. No values have been assumed for these items.

OPD-001 Notification Template Structure and Template ID Policy

Referenced from: Notification Matrix, Template ID column (§17.15).

Product Owner approval is required for:

- Template naming convention.
- Template identifier format.
- Required template variants.
- Channel-specific template handling.
- Localization and versioning rules, where applicable.

OPD-003 Document Governance Matrix

Referenced from: Document Matrix, Owning Entity / Authorized Roles / Default Classification / Retention Rule / Archive Rule columns (§18.21). A proposed matrix has been prepared for Product Owner review and is pending approval.

Product Owner approval is required for:

- Owning business entity, per document type.
- Authorized roles, per document type.
- Default classification, per document type.
- Retention rule, per document type.
- Archive rule, per document type.

The specification is usable as a developer baseline, but the Document Matrix cannot be considered fully implementation-complete until OPD-001 and OPD-003 are resolved and incorporated.